NUMERO DI NEPERO e

Funzione esponenziale

$y = a^x \qquad a > 0, \quad a \neq 1$

$y = 2^x$

$y = \left(\frac{1}{3}\right)^x$

$a = e$

$y = e^x \qquad e = 2{,}718\ldots$ numero irrazionale.

Montante dopo 1 anno con rendimento 100%

$$M = \underbrace{1000}_{C} + 1000 \cdot \underbrace{100\%}_{1} = 1000\left(1 + \underbrace{100\%}_{r}\right)$$

M = montante

C = capitale

r = rendimento

	r
100% →	1
50% →	0,5
25% →	0,25
1% →	0,01

$$M = C(1 + r)$$

Se la banca decide di dividere l'anno in 2 periodi (6 mesi) in ciascun periodo applica un tasso pari a $\frac{r}{2}$

$$M_1 = \left(1000 + 1000 \cdot 50\%\right) = 1500$$

dopo 6 mesi

$$M_2 = \left(1500 + 1500\,50\%\right) = 2250$$

$$= \left(1000 + 1000 \cdot 50\%\right) + \left(1000 + 1000 \cdot 50\%\right) \cdot 50\%$$

$$= \left(1000 + 1000 \cdot 50\%\right)\left(1 + 50\%\right)$$

$= 1000\left(1+50\%\right)\left(1+50\%\right) = 1000\left(1+50\%\right)^2$

$$M = C\left(1+\frac{r}{2}\right)^2$$

Se la banca decide di dividere l'anno in 3 periodi (4 mesi) e in ciascun periodo applica un tasso pari a $\frac{r}{3}$:

$$M = C\left(1+\frac{r}{3}\right)^3$$

$$M = 1000 \cdot \left(1+\frac{100\%}{3}\right)^3 = 2'370.$$

Se la banca paga gli interessi giornalmente:

$$M = C\left(1+\frac{r}{365}\right)^{365} =$$

$$M = 1000\left(1+\frac{100\%}{365}\right)^{365} = 2714$$

$2,714 \simeq e$

Im generale per n periodi: $M = C\left(1+\frac{r}{n}\right)^n$

$e = \left(1+\frac{1}{n}\right)^n$ per $n \to \infty$

Lunedì	Martedì	Giovedì	Venerdì	Sabato
I Mat		II Fis	II Mat	
II Fis	III Mat	III Mat		III Fis

RICHIAMO SULLE PROP. POTENZE

$a^n \cdot a^m = a^{n+m}$

$a^n \cdot b^n = (a \cdot b)^n$

$a^m + a^n = a^{n+m-n} + a^n = a^n \cdot a^{m-n} + a^n = a^n (a^{m-n} + 1)$

$3^5 + 3^2 = 3^2 \cdot 3^3 + 3^2 = 3^2 (3^3 + 1) = 28 \cdot 3^2$

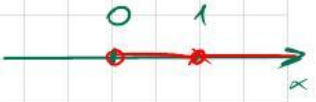
$9^5 + 3^3 = (3^2)^5 + 3^3 = 3^{10} + 3^3 = 3^3 \cdot 3^7 + 3^3 = 3^3 (3^7 + 1)$

$3^5 + 5^2$ la lascio così

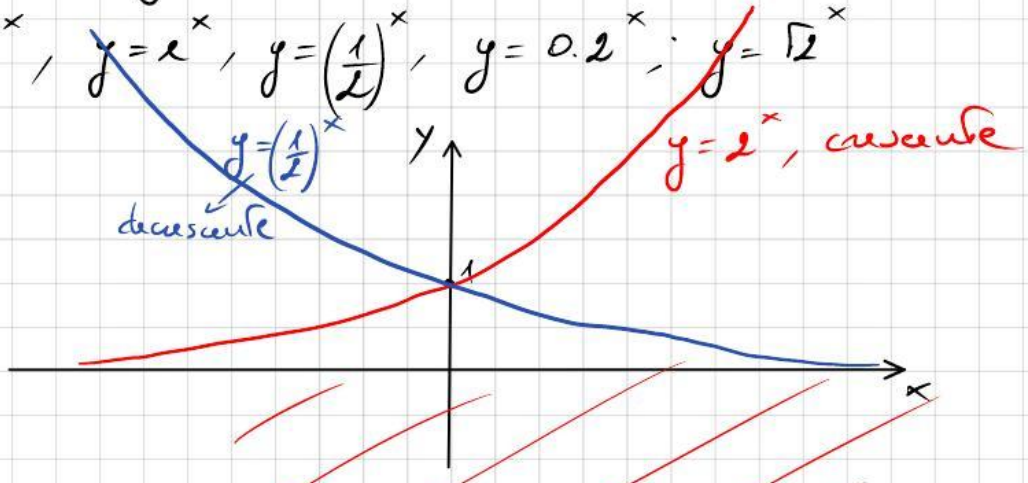
FUNZIONE ESPONENZIALE

Def:
Chiamiamo FUNZIONE ESPONENZIALE ogni funzione del tipo:

$$y = a^x \qquad a > 0,\ a \neq 1,\ \forall x \in \mathbb{R}$$



Es: $y = 2^x$, $y = e^x$, $y = \left(\frac{1}{2}\right)^x$, $y = 0.2^x$, $y = \sqrt{2}^x$



Tutte le funzioni esponenziali hanno l'asse x come asintoto.

$y = \left(\frac{1}{2}\right)^x = 2^{-x}$ $\quad 2^x$ e $\left(\frac{1}{2}\right)^x$ sono simmetriche rispetto all'asse y.

Tutte le funzioni esponenziali passano per il punto (0, 1)
Il dominio è $\mathbb{R}$.

FUNZIONI ESPONENZIALI NON ELEMENTARI $\quad y = [f(x)]^{g(x)}$

- $y = a^{f(x)}$

$y = 3^{\sqrt{x-1}}$ $\qquad$ Dominio: $x - 1 \geq 0$; $x \geq 1$

- $y = [f(x)]^{g(x)}$

$y = (4x^2 - 1)^{\sqrt{x}}$

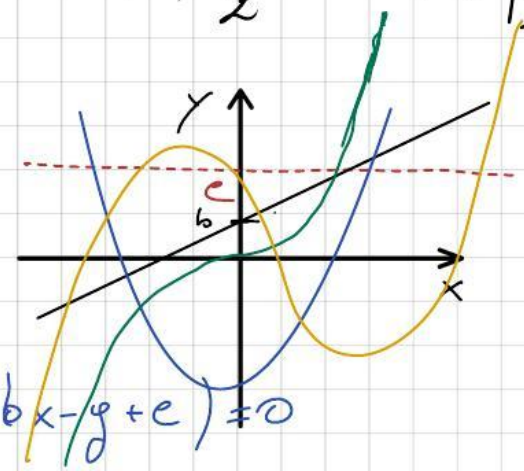
$(a > 0 \; ; \; a \neq 1)$

$\begin{cases} 4x^2 - 1 > 0 \rightarrow x < -\frac{1}{2} \vee x > \frac{1}{2} \\ 4x^2 - 1 \neq 1 \rightarrow 4x^2 \neq 2 \rightarrow x^2 \neq \frac{1}{2} \rightarrow x \neq \pm\sqrt{\frac{1}{2}} \\ x \geq 0 \end{cases}$

$x > \frac{1}{2}$

FUNZ.

$y = c + 0 \cdot x$



$y = ax + b$ retta

$y = ax^2 + bx + c$ $\quad (ax^2 + bx - y + c) \neq 0$

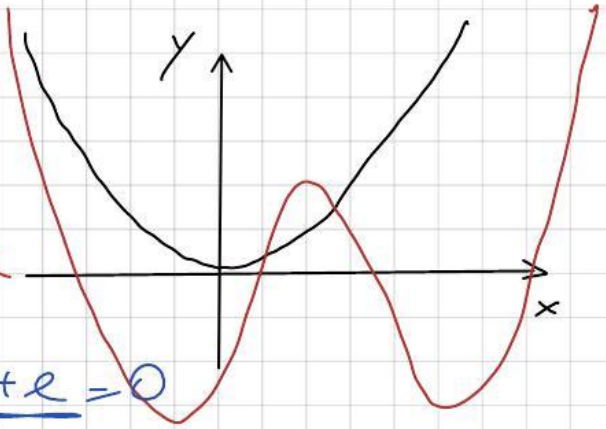
$y = x^3$

$y = ax^3 + bx^2 + cx + d$

$y = x^4$

$y = ax^4 + bx^3 + cx^2 + dx + e$

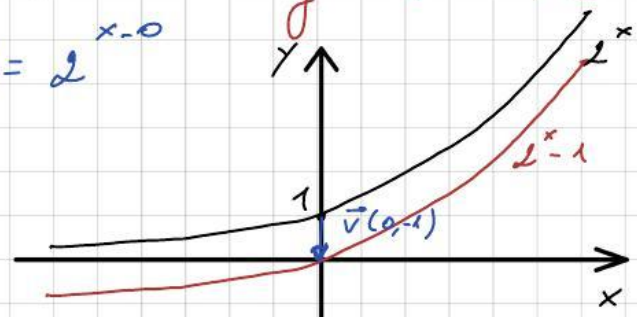
x	y
-3	-27
-2	-8
-1	-1
0	0
1	1
2	8
3	27



$ax^2 + by^2 + cx + dy + e = 0$

GRAFICO DI $y = a^{(x-c)} + b$

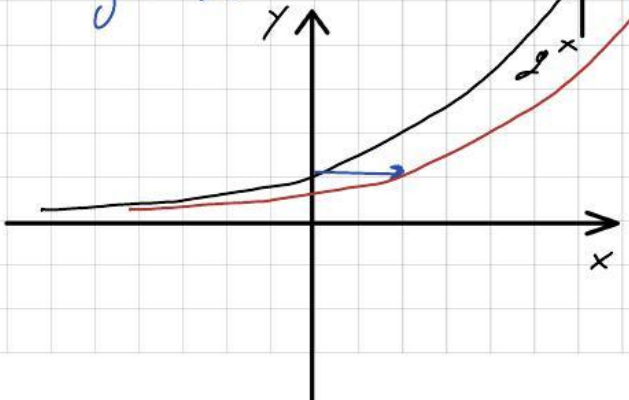
$y = 2^x$ $\quad y - (-1) = 2^{x-0}$

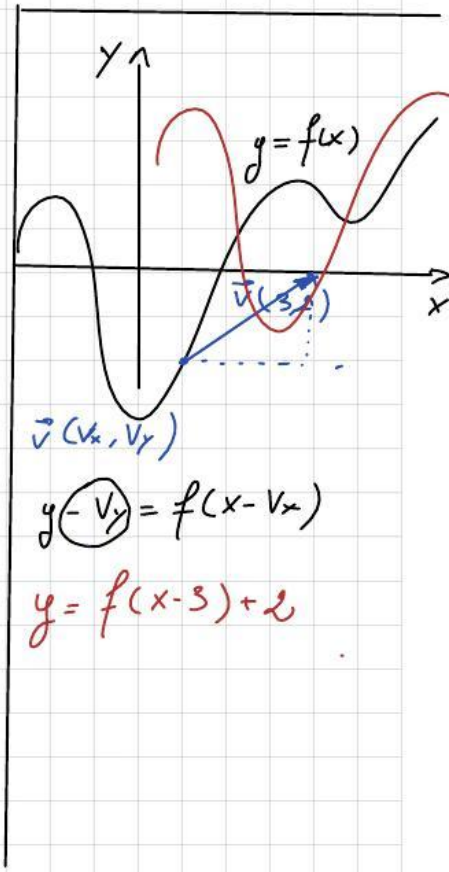


$y = 2^x - 1$

$y + 1 = 2^x$

$y = 2^{x-2}$ $\quad y - 0 = 2^{x-2}$



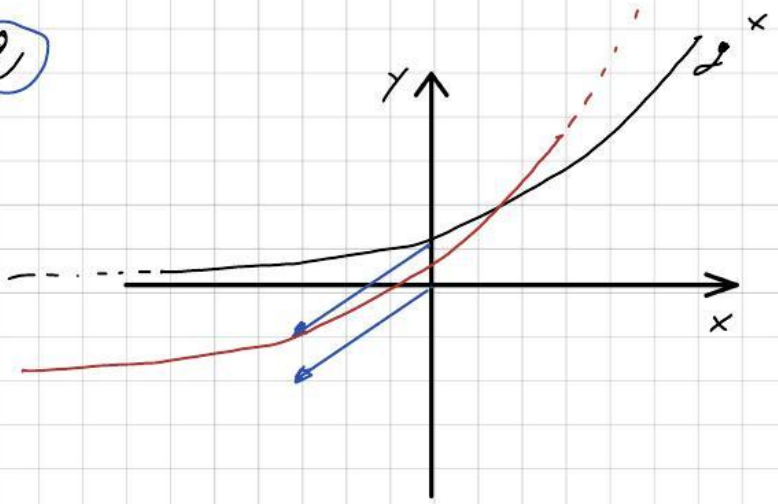


$y - v_y = f(x - v_x)$

$y = f(x-3) + 2$

$y = 2^{x+3} - 2$

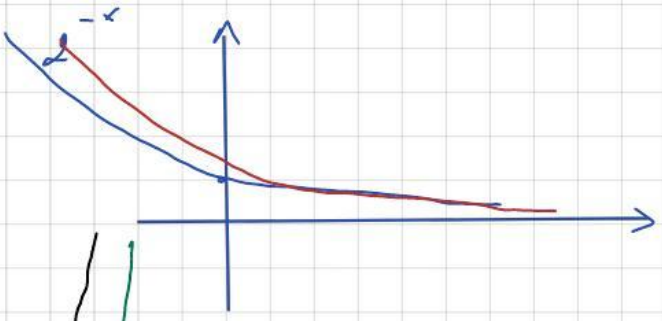
$V(-3, -2)$



$y = 2^{-x+1}$

$y = 2^{-(x-1)}$

$y = 2^{-x}$



n° 113

$y = |1 - 4^x| - 1$

$y = 4^x$

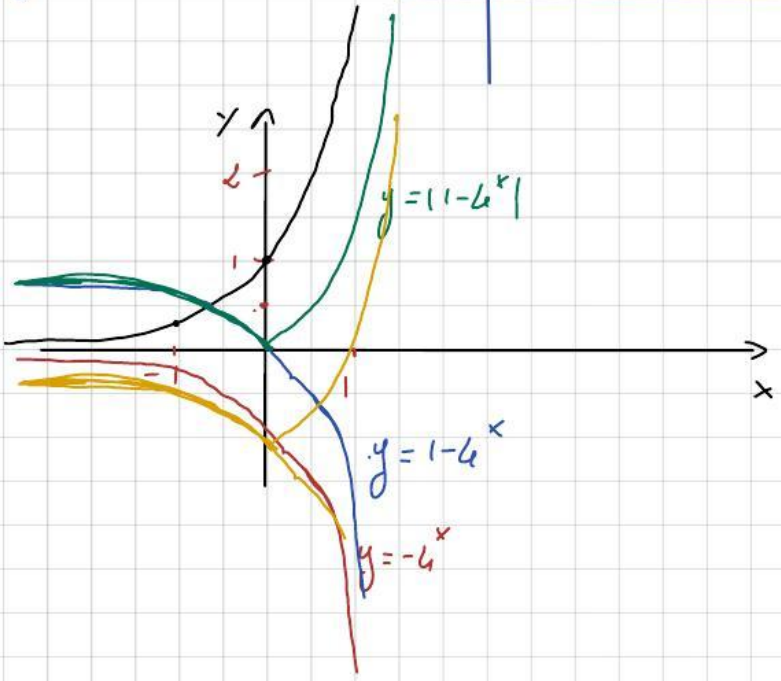
$y = -4^x$

$y = 1 - 4^x = -4^x + 1$

$y = |1 - 4^x|$

$y = |1 - 4^x| - 1$

x	y
-1	1/4
0	1
1	4
2	16



EQUAZIONI ESPONENZIALI

Un'equazione si dice ESPONENZIALE se contiene almeno una potenza con l'incognita nell'esponente

$$a^x = b \qquad a > 0$$

- Se $a = 1$

  $1^x = b \rightarrow b = 1$

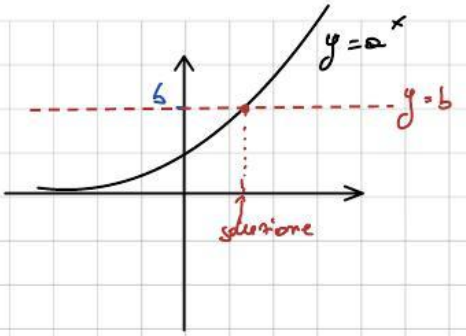
- se $b \leq 0 \rightarrow a^x = b$ impossibile

- se $b > 0 \rightarrow a^x = b$ ha sempre una unica soluzione.

$a^x = b$

$\begin{cases} y = a^x \\ y = b \end{cases}$ Troviamo il punto di intersezione tra le 2 funzioni.

$a^x = b$



Esempio:

$$25^x = 125$$

$$(5^2)^x = 5^3$$

$$5^{2 \cdot x} = 5^3$$

Due potenze sono uguali se hanno la stessa base e lo stesso esponente

Pongo l'uguaglianza tra gli esponenti

$$2x = 3$$

$$x = \frac{3}{2}$$

n° 163

- $2^{3x-1} = 16$

$$2^{3x-1} = 2^4 \rightarrow 3x - 1 = 4 \rightarrow x = \frac{5}{3}$$

n° 171

- $2^x + 2^{x+1} = -2^{x-1} + 7$

$$2^x + 2^{x+1} + 2^{x-1} = 7$$

$$2^{x-1}(2 + 4 + 1) = 7$$

$7 \cdot 2^{x-1} = 7$ $\quad 2^{x-1} = 1 \rightarrow 2^{x-1} = 2^0$

$$x - 1 = 0$$

$$x = 1$$

n° 210

- $3 \cdot 36^x - 6 \cdot 3^x - 4^x + 1 = 0$

- Raccogli parzialmente i primi 2 e i secondi 2
- Scomponi in fattori
- Risolvi applicando la legge dell'annullamento del prodotto.

$3 \cdot [3^2 2^2]^x - 6 \cdot (3^2)^x - (2^2)^x + 1 = 0$

$3 \cdot 3^{2x} \cdot 2^{2x} - \frac{3 \cdot 2}{6} \cdot 3^{2x} - 2^{2x} + 1 = 0$

$3 \cdot 3^{2x}(2^{2x} - 1) + (2^{2x} - 1) = 0$

$(2^{2x} - 1) \cdot (3 \cdot 3^{2x} - 1) = 0$

$2^{2x} - 2 = 0$

$2^{2x} = 2^1$

$2x = 1$

$x = \frac{1}{2}$

$3 \cdot 3^{2x} - 1 = 0$

$3^{2x+1} = 1 = 3^0$

$2x + 1 = 0$

$2x = -1$

$x = -\frac{1}{2}$

$S = \left\{ \pm \frac{1}{2} \right\}$

n° 152

$\begin{cases} 2x + y = 3 \\ 2^{x-y} = 64 \end{cases} \quad \begin{cases} y = 3 - 2x \\ 2^{x-(3-2x)} = 64 \end{cases} \quad \begin{cases} = \\ 2^{x-3+2x} = 64 \end{cases} \quad \begin{cases} = \\ 2^{3x-3} = 2^6 \end{cases} \quad \begin{cases} y = 3 - 2x \\ 3x - 3 = 6 \end{cases} \quad \begin{cases} y = 3 - 2 \cdot 3 = -3 \\ x = 3 \end{cases}$

E> pg 613 n° 143, 144, 145, 156, 161, 164, 211, 214, 255

n° 185 $4^x = 2^x - 2 \rightarrow (2^x)^2 - 2^x + 2 = 0$ $\quad 2^x = t$

$(2^2)^x = (2^x)^2$

$t^2 - t + 2 = 0$

$(t-2)\cdot(t+1) = 0$

$t = \boxed{2^1 = 2^x} \rightarrow x = 1$

$t = \boxed{-1 = 2^x}$ imp.

n° 199

$3^x + 3^{1-x} = 4$

n° 205

$2^{4x+3} + 2 = 17 \cdot 4^x$

pg 616 dal n° 230 ÷ 237

251 ÷ 256

$e = 2,718281828459045$

2 7 1 8 2 8 1 8 2 8 1 8

Ai modesti o vanitosi ai vidusi o timorosi

do 8 cantato 4 gaio 5 ritmo 3 logaritmo 2

$|8^x - 2| = \sqrt{2^{3x}}$

- $8^x - 2 = \sqrt{2^{3x}}$

$8^x - 2 = 2^{\frac{3}{2}x}$

$2^{3x} - 2 = 2^{\frac{3}{2}x}$

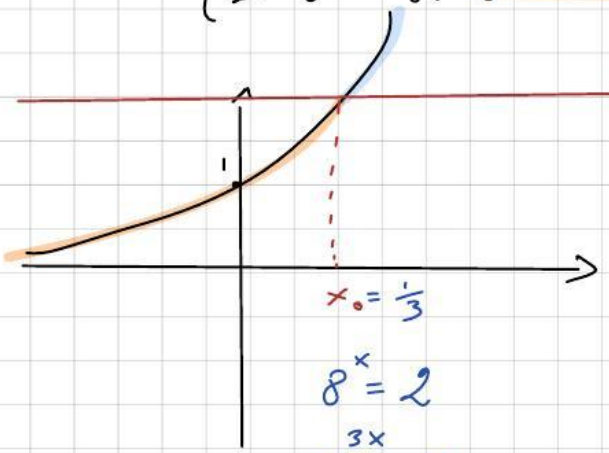
$\rightarrow (2^{3x})^2 - 2^{\frac{3x}{2}} - 2 = 0$ $\quad t = 2^{\frac{3x}{2}}$

$T^2 - T - 2 = 0$

$(T+1)(T-2) = 0$ $\quad T = -1$ inf. $\quad T = 2^1 = 2^{\frac{3x}{2}}$

$\frac{3x}{2} = 1 \quad x_1 = \frac{2}{3}$

$|8^x - 2| = \begin{cases} 8^x - 2 & \text{se } 8^x - 2 \geq 0,\ x \geq \frac{1}{3} \\ 2 - 8^x & \text{se } 8^x - 2 < 0,\ x < \frac{1}{3} \end{cases}$



$8^x = 2$

$2^{3x} = 2$

$x = \frac{1}{3}$

- $2 - 8^x = \sqrt{2^{3x}}$

$2 - (2^{3x})^2 = 2^{\frac{3x}{2}}$ $\quad T^2 \quad T$

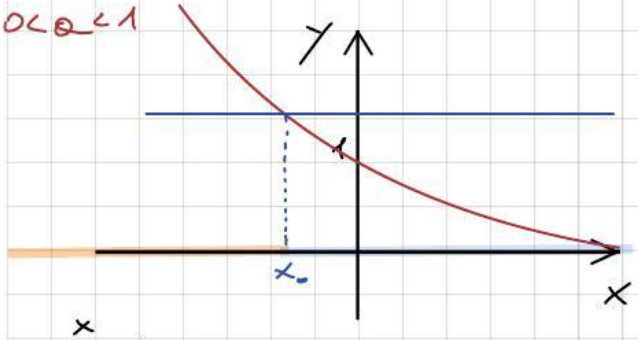
$T^2 + T - 2 = 0$

$(T+2)(T-1) = 0$ $\quad T = -2$ $\quad T = 1 = 2^{\frac{3x}{2}}$

$2^0 = 2^{\frac{3x}{2}}$ $\quad \frac{3x}{2} = 0 \rightarrow x_2 = 0.$

DISEQ. ESPONENZIALI

una disequazione esponenziale contiene almeno una potenza con l'incognita nell'esponente



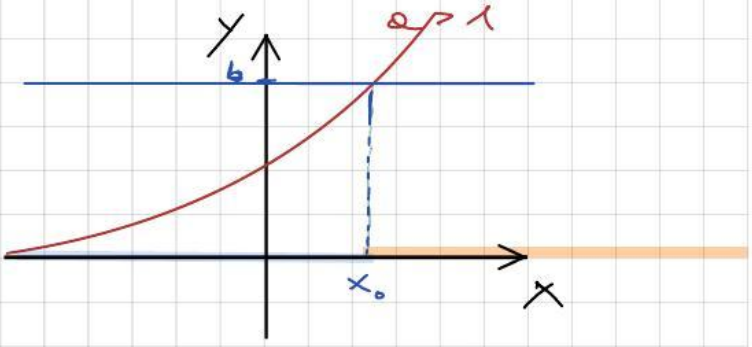
- $a^x > b$

Risolvo l'eq. associata

$a^x = b \rightarrow x = x_0$

La disq. data ha soluzione

$x < x_0$



- $a^x > b$ con $b > 0$

e funzioni

$y = a^x, \quad y = b$

Risolvo l'eq. associata $a^x = b \rightarrow x = x_0$

La disequazione data ha soluzione

$x > x_0$

- $a^x < b$

Risolvo l'eq. associata

$a^x = b \rightarrow x = x_0$

La disq. data ha soluzione

$x > x_0$

- $a^x < b$

Risolvo $a^x = b \rightarrow x = x_0$

La disq. data ha soluzione

$x < x_0$

Esercizio:

$32^x > 128$

$2^{5 \cdot x} > 2^7$

$\left(2^{5x} = 2^7 \; ; \quad 5x = 7 \; ; \quad x = \frac{7}{5} \right)$

$x > \frac{7}{5}$

$(x+1) \cdot 2^x > 4$

$x + 1 > \frac{2^2}{2^x}$

$\underbrace{x+1}_{g=x+1} > \underbrace{2^{2-x}}_{f=}$

$2^{2-x} + \frac{3}{4} x^2 > 4$

$2^{2-x} > 4 - \frac{3}{4} x^2$

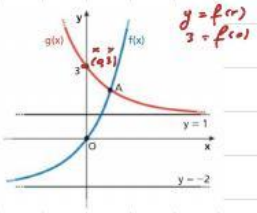
53 **LEGGI IL GRAFICO**

a. Trova, deducendo i dati dalla figura, le espressioni delle funzioni $f(x) = b \cdot 2^x + c$ e $g(x) = k \cdot 2^{-x} + h$ rappresentate e determina le coordinate del punto di intersezione A dei loro grafici.

b. Determina la retta appartenente al semipiano $x < 0$ e parallela all'asse y che, intersecando $f(x)$ e $g(x)$, forma un segmento BC lungo $\frac{21}{2}$.

c. Calcola il perimetro del triangolo ABC.

[a) $f(x) = 2^{x+1} - 2$, $g(x) = 2^{-x+1} + 1$, $A(1; 2)$; b) $x = -2$; c) $\frac{21 + \sqrt{85}}{2} + \sqrt{58}$]



$y = f(x)$
$3 = f(a)$

$y = a^x$

$y = a^{x-2}$

$f(x) = b \cdot 2^x + c$ → c rappresenta la traslazione lungo la direzione y

$g(x) = k \cdot 2^{-x} + h$ → la funzione è traslata di 1 verso l'alto → $h = 1$

Imponiamo il passaggio di $f(x)$ dall'origine $(0, 0)$

$f(0) = 0$

$0 = b \cdot 2^0 + c \rightarrow b + c = 0 \rightarrow b = 2$

$c = -2$

$f(x) = 2 \cdot 2^x - 2 = 2^{x+1} - 2$

Imponiamo il passaggio di $g(x)$ per il punto $(0,3)$

$g(0)=3$

$3 = K \cdot 2^{0} + 1 \rightarrow K=2$

$g(x) = \overset{K}{2} \cdot 2^{-x} + \overset{h}{1} = 2^{1-x} + 1$

Troviamo A

$f(x)=g(x)$

$2^{x+1} - 2 = 2^{-x+1} + 1$

$2^{x+1} = 2^{-x+1} + 3 \quad ; \quad 2 \cdot 2^{x} = 2 \cdot 2^{-x} + 3$ (t, $\frac{1}{t}$)

$2t = \frac{2}{t} + 3 \rightarrow 2t^2 - 3t - 2 = 0$

$t_{1,2} = \frac{3 \pm \sqrt{9+16}}{4} = \frac{3 \pm 5}{4}$ $\quad t=2 \quad -\frac{2}{2}=2$

$2^{x} = 2 \rightarrow x=1$

$g = 2^{1+1} - 2 = 2$

$A(1,2)$.

54 Considerata la funzione $f(x) = a2^{x} + b2^{-x} + c$,

a. determina a, b, c in modo che il suo grafico sia simmetrico rispetto all'asse y, passi per $\left(1; \frac{7}{2}\right)$ e $f(0) = 4$;

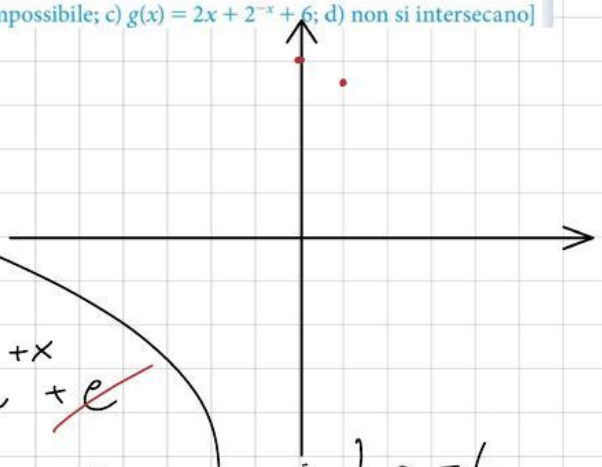
b. calcola per quali valori di x è $f(x) \geq \frac{9}{2}$;

c. esprimi analiticamente la funzione $g(x)$ il cui grafico è simmetrico di quello di $f(x)$ rispetto alla retta di equazione $y = 6$;

d. trova le intersezioni dei grafici di $f(x)$ e $g(x)$.

[a) $a = b = -1$, $c = 6$; b) impossibile; c) $g(x) = 2x + 2^{-x} + 6$; d) non si intersecano]

$\begin{cases} f(x) = f(-x) \\ f(1) = \frac{7}{2} \\ f(0) = 4 \end{cases}$



$a2^{x} + b2^{-x} + c = a2^{-x} + b2^{+x} + c$

$2^{x}(a-b) + 2^{-x}(-b+a) = 0$

$(a-b)(2^{x} - 2^{-x}) = 0$

$\begin{cases} a=b \\ 2^{x} = 2^{-x} \end{cases}$

$\begin{cases} a=b \\ 2b + \frac{b}{2} + c = \frac{7}{2} \\ 2b + c = 4 \end{cases}$

$\begin{cases} 2b + \frac{b}{2} + 4 - 2b = \frac{7}{2} \\ c = 4 - 2b \end{cases}$

$\begin{cases} a = -1 \\ b = 7 - 8 \; ; \; b = -1 \\ c = 4 - 2(-1) = 6 \end{cases}$

$\begin{cases} a=b \\ 2a + \frac{1}{2}b + c = \frac{7}{2} \\ a + b + c = 4 \end{cases}$

$y = -2^{x} - 2^{-x} + 5$

DEFINIZIONE DI LOGARITMO

DEFINIZIONE

Dati due numeri reali positivi a e b, con $a \neq 1$, chiamiamo **logaritmo in base a di b** l'esponente x da assegnare alla base a per ottenere il numero b.

$$\log_a b = x \leftrightarrow a^x = b$$
$$a > 0,\ a \neq 1,\ b > 0$$

$3^2 = 9 \iff \log_3 9 = 2$ (9 = argomento del logaritmo; 3 = base del logaritmo)

$\log_a b$ ($b > 0$)

$a > 0,\ a \neq 1$

$\log_a x = \log_a y \Rightarrow x = y$

- $\log_2 16 = 4$
- $\log_{\frac{1}{3}} 9 = -2$ $\left(\left(\frac{1}{3}\right)^{-2} = 9\right)$
- $\log_3 3 = 1$
- $\log_4 2 = \frac{1}{2}$
- $\log_5 1 = 0$
- $\log_3 3^3 = 3$ $\quad 3^3 = 3^3$
- $\log_3 3^4 = 4$
- $4 \cdot \log_3 3 = 4$ ($\log_3 3 = 1$)

PROPRIETÀ DEI LOGARITMI

Prodotto: $\log_a (b \cdot c) = \log_a b + \log_a c$ se $b > 0, c > 0$

Quoziente: $\log_a \left(\frac{b}{c}\right) = \log_a b - \log_a c$ " "

Potenza: $\log_a b^n = n \cdot \log_a b$

Cambiamento di base: $\log_a b = \frac{\log_c b}{\log_c a}$

Dimostriamo la 1ª proprietà:

$x = \log_a b \quad ; \quad y = \log_a c$

$a^x = b \qquad a^y = c$ moltiplichiamo m.a.m

$a^x \cdot a^y = b \cdot c$

$a^{x+y} = b \cdot c$

$\log_a (b \cdot c) = x + y$

$$\log_a (b \cdot c) = \log_a b + \log_a c$$

25 $\log_3 \frac{3 \cdot \sqrt{3}}{\sqrt[3]{9}}$; $\log_5 \left(0{,}2 \frac{\sqrt{5}}{5}\right)$.

26 $\log_3 (27 \cdot \sqrt{3})$; $\log_4 \frac{1}{2}$.

27 $\log_{625} \frac{\sqrt[3]{5}}{25}$; $\log_{16} \frac{2}{\sqrt[7]{2}}$.

$$\log_{\frac{1}{2}} \frac{\sqrt{2}}{2} = \log_{\frac{1}{2}} \frac{\sqrt{2}}{\sqrt{2} \cdot \sqrt{2}} = \log_{\frac{1}{2}} \frac{1}{\sqrt{2}} = \log_{\frac{1}{2}} \sqrt{\frac{1}{2}} = \log_{\frac{1}{2}} \left(\frac{1}{2}\right)^{\frac{1}{2}} = \frac{1}{2}$$

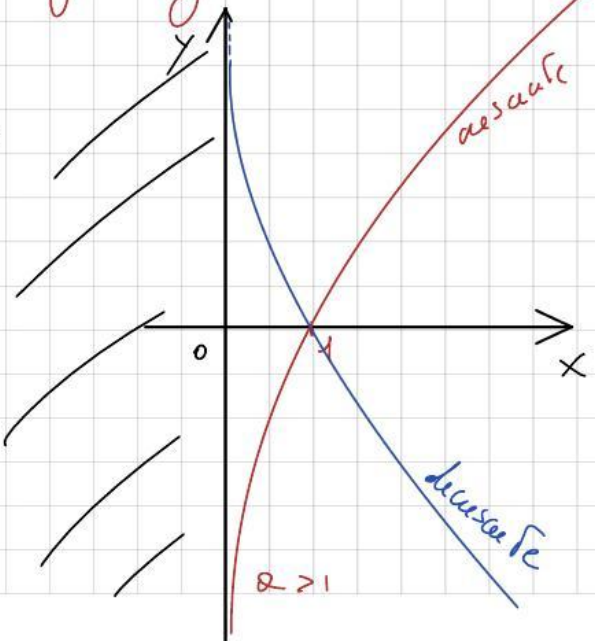
LA FUNZIONE LOGARITMO

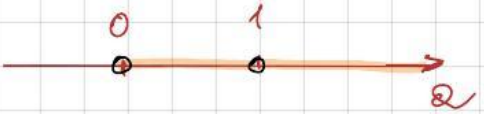
Una funzione logaritmica è del tipo:

$y = \log_a x \qquad a > 0, \ a \neq 1$

Dominio: $x > 0$

L'asse y è asintoto per la funzione logaritmo.



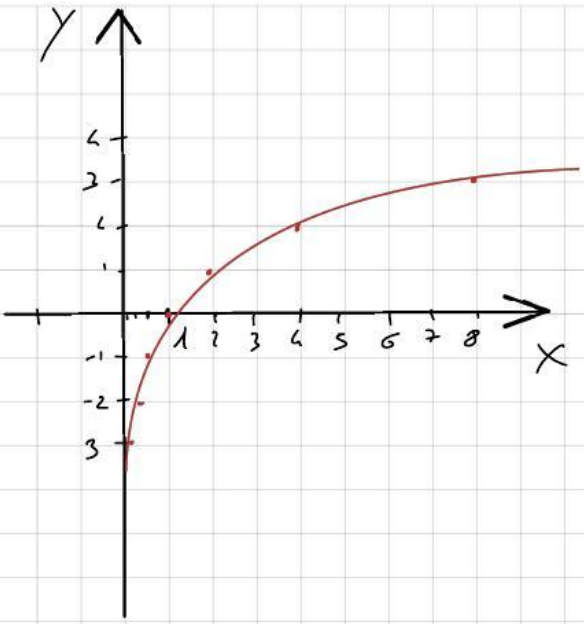


$\log x = \log_{10} x$

$\ln x = \log_e x$

Tracciare la funzione $y = \log_2 x$

x	y	
$\frac{1}{8}$	$\log_2 \frac{1}{8} = -3$	$\left(2^{-3} = \frac{1}{8}\right)$
$\frac{1}{4}$	$\log_2 \frac{1}{4} = -2$	
$\frac{1}{2}$	$\log_2 \frac{1}{2} = -1$	
1	$\log_2 1 = 0$	
2	$\log_2 2 = 1$	
4	$\log_2 4 = 2$	
8	$\log_2 8 = 3$	



EQUAZIONI LOGARITMICHE

Un'equazione logaritmica è l'incognita che compare nell'argomento di almeno un logaritmo.

$$\log_a A(x) = \log_a B(x) \Rightarrow A(x) = B(x)$$

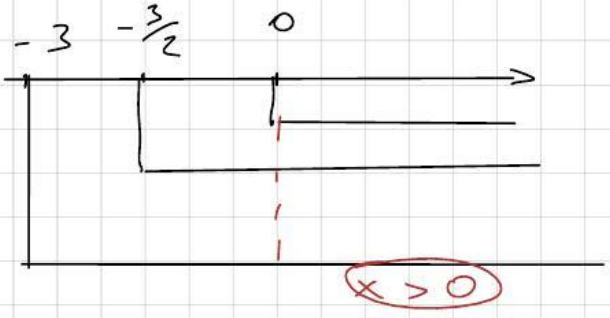
Due logaritmi sono uguali se hanno la stessa base e lo stesso argomento

PRIMA DI RISOLVERE L'EQUAZIONE BISOGNA PORRE LE C.E.

$$\log x + \log(x+3) = \log 2 + \log(2x+3)$$

C.E.:

$$\begin{cases} x > 0 & x > 0 \\ x+3 > 0 & x > -3 \\ 2x+3 > 0 & x > -\frac{3}{2} \end{cases}$$



$$\log_{10}(x(x+3)) = \log_{10}(2\cdot(2x+3))$$

Poniamo l'uguaglianza tra gli argomenti:

$$x(x+3) = 2(2x+3)$$

$$x^2 + 3x - 4x - 6 = 0$$

$$x^2 - x - 6 = 0$$

$$(x+2)(x-3) = 0 \quad \begin{cases} x_1 = -2 \\ x_2 = 3 \end{cases}$$

non accettabile

$x = 3$

pag 667

Es n° 283 e 300, 316, 315, 323, 358.

$$\log_2(x^2+1) = 1 + \frac{2}{3}\log_2 x + \log_8 x$$

$$\log_a b = \frac{\log_c b}{\log_c a}$$

$$\log_8 x = \frac{\log_2 x}{\log_2 8} = \frac{\log_2 x}{3} = \frac{1}{3}\log_2 x$$

$$\log_2(x^2+1) = \log_2 2 + \frac{2}{3}\log_2 x + \frac{1}{3}\log_2 x$$

$$\log_2(x^2+1) = \log_2 2 + \left(\frac{2}{3}+\frac{1}{3}\right)\log_2 x$$

$$\log_2(x^2+1) = \log_2 2 + \log_2 x$$

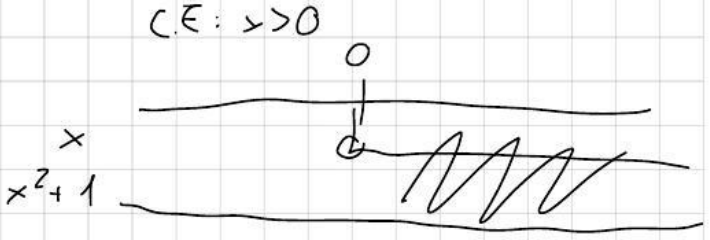
$$\log_2(x^2+1) = \log_2(2\cdot x)$$

$$x^2 + 1 = 2x$$

$$x^2 - 2x + 1 = 0$$

$$(x-1)^2 = 0 \iff x = 1$$

C.E: $x > 0$



$$\log_{a^n} b = \frac{\log_a b}{\log_a a^n} = \frac{\log_a b}{n} = \frac{1}{n}\cdot\log_a b$$

Es: $\log_8 x = \log_{2^3} x = \frac{1}{3}\log_8 x$

Es: $\log_{\frac{1}{2}} x = \log_{2^{-1}} x = \frac{\log_2 x}{-1} = -\log_2 x$

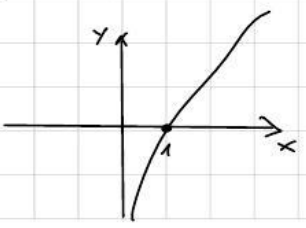
$$3\log^2 x - 2\log x = 0 \qquad (t^2 \to \log^2 x,\ t \to \log x)$$

$$\log^2 x = (\log x)^2 = \log x \cdot \log x$$

$$3t^2 - 2t = 0$$

C.E. $x > 0$

$$t(3t-2) = 0 \quad \begin{cases} t = 0 = \log x \to x = 1 \\ t = \frac{2}{3} = \log x \to \end{cases}$$



$x = \log_a a^x$

$3 = \log_5 5^3$

$3 = \log 10^3$

$3 = \log_{\frac{1}{2}} \left(\frac{1}{2}\right)^3$

$\log_{10} 10^{\frac{2}{3}} = \log_{10} x$

$10^{\frac{2}{3}} = x$

$x = \sqrt[3]{100}$

$1 = \frac{3}{2} \log x$

$\log 10 = \log x^{\frac{3}{2}}$

$x^{\frac{3}{2}} = 10$

$x = \sqrt[3]{100}$

n° 330

$1 - \ln^2 x = 0$

$1 - (\ln x)^2 = 0$

$t = \ln x$

$1 - t^2 = 0$

$t^2 = 1$

$t = \pm 1$

$t^2 - 1 = 0$

$(t-1)(t+1) = 0$

$t_1 = 1$

$t_2 = -1$

$t_1 = \ln x \rightarrow \log_e x = 1$

$e^1 = x \rightarrow x = e$

$t_2 = \ln x \rightarrow \log_e x = -1$

$e^{-1} = x \rightarrow x = \frac{1}{e}$

$\ln x = \log_e x$

$\log x = \log_{10} x$

$\ln x = -\ln e$

$\ln x = \ln e^{-1}$

$x = e^{-1} = \frac{1}{e}$

$\ln x = \log_e x$

n° 379

$$\frac{\log_2^2 x}{\log_2 x + 3} + \frac{6}{\log_2 x - 3} + \frac{72}{3 - \log_2^2 x} = 0$$

$R = \left\{\frac{1}{512}, 64\right\}$

C.E. $x > 0$

C.E.

$$\begin{cases} x > 0 \\ \log_2 x + 3 \neq 0 \\ \log_2 x - 3 \neq 0 \end{cases}$$

$t = \log_2 x$

$$\frac{t}{t+3} + \frac{6}{t-3} - \frac{72}{\log_2 x^2 - 9} = 0$$

$$\frac{t}{t+3} + \frac{6}{t-3} - \frac{72}{t^2 - 9} = 0$$

$$\frac{t}{t+3} + \frac{6}{t-3} - \frac{72}{(t+3)(t-3)} = 0$$

$(t-3)t + (t+3)6 - 72 = 0$

$$\begin{cases} \log_2 x \neq -3 \\ \log_2 x \neq 3 \end{cases} \quad \begin{cases} x > 0 \\ 2^{-3} \neq x \\ 2^3 \neq x \end{cases}$$

$t^2 - 3t + 6t + 18 - 72 = 0$

$t^2 + 3t - 54 = 0 \qquad \Delta = 15^2 \qquad t_{1,2} = \frac{-3 \pm 15}{2} \begin{cases} 6 \\ -9 \end{cases}$

$\log_2 x = 6 \qquad \log_2 x = -9$

$2^6 = 64 \qquad 2^{-9} = \frac{1}{512}$

n° 383 $\qquad x > 0$

$$\sqrt{\log_2^2 x + \log_2 x - 2} = \log_{\frac{1}{2}} x - 2$$

$A(x)$

$$\left(\begin{cases} \sqrt{A(x)} = B(x) \\ \begin{cases} A(x) \geq 0 \\ B(x) \geq 0 \\ A(x) = [B(x)]^2 \end{cases} \end{cases} \Rightarrow \begin{cases} B(x) \geq 0 \\ A(x) = [B(x)]^2 \end{cases} \right)$$

≥ 0

$$B(x) \geq 0 \rightarrow \log_{\frac{1}{2}} x - 2 \geq 0$$

$$A(x) = [B(x)]^2 \rightarrow \log_2^2 x + \log_2 x - 2 = \left(\log_{\frac{1}{2}} x - 2 \right)^2$$

$\vdots$

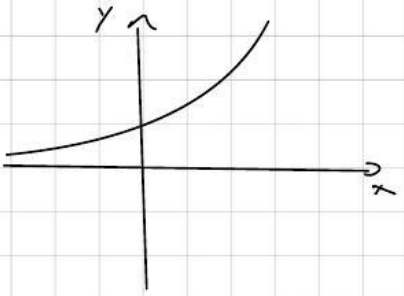
Es pag 465 n° 383, 356, 357, 355, 384, 386

$$\log_3 x = \frac{\log_5 x}{\log_5 3} \; ; \; \log_{27} x = \frac{\log_3 x}{\log_3 27} \quad (\log_3 27 = 3)$$

$$\log_{a^n} x = \frac{\log_a x}{\log_a a^n} = \frac{\log_a x}{n}$$

$$\log_{a^\mu} x = \frac{1}{\mu} \cdot \log_a x$$

$$\log_a x^\mu = \mu \log_a x$$



384 $\log_{\frac{1}{4}}\left(-\frac{1}{3^{2-x}}+3^x\right)=-1+\log_{\frac{1}{4}}(2^{x-3}+2^x)$

C.E. $\begin{cases} -\frac{1}{3^{2-x}}+3^x>0 \\ 2^{x-3}+2^x>0 \end{cases}$ $\forall x\in\mathbb{R}$ (somma di quantità positive)

$-3^{x-2}+3^x>0$

$-\frac{3^x}{9}+3^x>0 \qquad 3^x\left(1-\frac{1}{9}\right)>0 \qquad \frac{8}{9}3^x>0 \quad \forall x\in\mathbb{R}$ ($3^x>0$)

$2^x\left(\frac{1}{8}+1\right)=\frac{9}{8}2^x$

$\log_{\frac{1}{4}}\left(\frac{8}{9}3^x\right)=\log_{\frac{1}{4}}4+\log_{\frac{1}{4}}\left(\frac{2}{8}+2^x\right)$

$\log_{\frac{1}{4}}\left(\frac{8}{9}3^x\right)=\log_{\frac{1}{4}}\left(4\cdot\frac{9}{8}2^x\right)$

$\frac{8}{9}3^x=\frac{9}{2}2^x \rightarrow \frac{3^x}{2^x}=\frac{9^2}{2^4} \rightarrow \left(\frac{3}{2}\right)^x=\frac{3^4}{2^4}$

$\left(\frac{3}{2}\right)^x=\left(\frac{3}{2}\right)^4 \rightarrow x=4$

$f(x) \quad g(x)$
$a = a$

359 $1-\frac{2}{\log_3 x+2}=3\log_{\frac{1}{3}} x$ $\left[\frac{\sqrt[3]{9}}{27};1\right]$

C.E. $\begin{cases} x>0 \\ \log_3 x+2\neq 0 \rightarrow \log_3 x\neq -2 \rightarrow 3^{-2}\neq x \rightarrow x\neq\frac{1}{9} \end{cases}$

$t=\log_3 x$

$\frac{\log_3 x+2-2}{\log_3 x+2}=\frac{3\left(-\log_3 x\right)\cdot(\log_3 x+2)}{\log_3 x+2}$ ($\log_{\frac{1}{3}} x=-\log_3 x$, t)

$\frac{t}{t+2}=\frac{-3t(t+2)}{t+2}$

$\log_{\frac{1}{a}} x=-\log_a x$

$3t^2+6t+t=0 \; ; \; 3t^2+7t=0 \; ; \; t(3t+7)=0$

$t=0=\log_3 x \; ; \; x=1$

$t=-\frac{7}{3}=\log_3 x \; ; \; 3^{-\frac{7}{3}}=x$

$x=\frac{1}{3^2\cdot 3^{\frac{1}{3}}}=\frac{1}{9\sqrt[3]{3}}=\frac{1}{9}\frac{1}{\sqrt[3]{3}}\frac{\sqrt[3]{3^2}}{\sqrt[3]{3^2}}=\frac{1}{9}\frac{\sqrt[3]{3^2}}{3}=\frac{1}{27}\sqrt[3]{9}$

$$a^{\frac{m}{n}} = a^{\frac{R}{D}} \cdot a \qquad \text{se } m > n$$

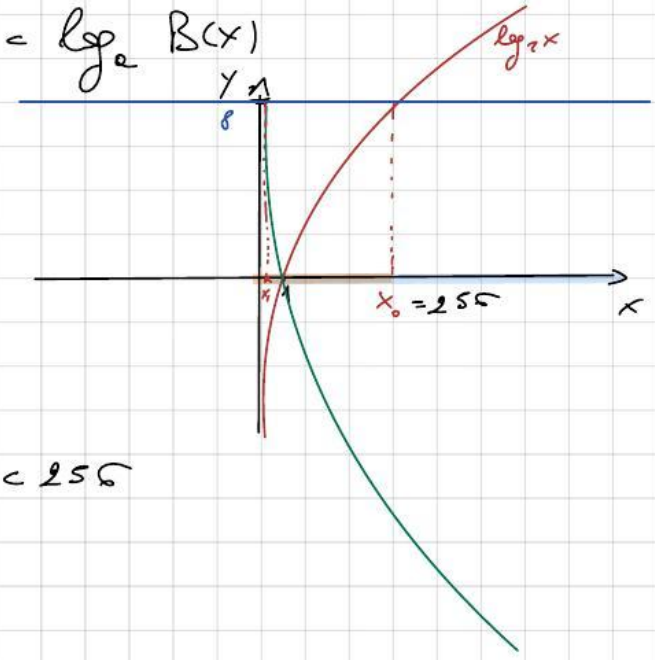
$17 : 5 = 3 \quad R = 2$
Div Divis Q.

$$a^{\frac{17}{5}} = a^{3} \cdot a^{\frac{2}{5}}$$

DISEQUAZIONI LOGARITMICHE pag 414.

una disequazione logaritmica ha l'incognita che compare nell'argomento di almeno un logaritmo

$$\log_a A(x) < \log_a B(x)$$



- $B(x) = 6$

$\log_2 x < 8$ C.E. $x > 0$

Risolvo l'eq. associata

$\log_2 x = 8 \rightarrow x = 2^8$

$x < 256$ S: $0 < x < 256$

$\log_{\frac{1}{2}} x < 8$

l'eq. associata: $\log_{\frac{1}{2}} x = 8 \rightarrow \left(\frac{1}{2}\right)^8 = x \Rightarrow x = \frac{1}{256}$

S: $x > \frac{1}{256}$

$$\log_a A(x) > \log_a B(x)$$

$a > 1$

C.E. $\begin{cases} \cancel{A(x) > 0} \\ B(x) > 0 \\ A(x) > B(x) \end{cases}$

$0 < a < 1$

C.E. $\begin{cases} A(x) > 0 \\ \cancel{B(x) > 0} \\ A(x) < B(x) \end{cases}$

n° 445

$$\frac{1}{2}\log(-x^2+2x) < \log x$$

$$\log(-x^2+2x) < 2\log x$$

C.E.
$x > 0$
$-x^2+2x > 0 \; ; \; x(x-2) < 0$

S: $0 < x < 2$

$\log_{10>1}(-x^2+2x) < \log_{10>1} x^2$

$-x^2+2x < x^2$

$2x^2-2x>0$

$x^2-x>0 \qquad x(x-1)>0$



$S: 1<x<2$

$x<0 \quad \vee \quad x>1$

LOGARITMI CON EQ. E DISEQ. ESPONENZIALI

$7 \cdot 5^{2x} = 3^{x+1}$

$\log(7 \cdot 5^{2x}) = \log 3^{x+1}$

$\log 7 + \log 5^{2x} = \log 3^{x+1}$

$\log 7 + 2x \log 5 = (x+1)\log 3$ ($x\log 3 + \log 3$)

$2x \log 5 - x \log 3 = -\log 7 + \log 3$

$x(2\log 5 - \log 3) = \log 3 - \log 7$

$x = \frac{\log 3 - \log 7}{2\log 5 - \log 3} \simeq -0{,}4$

$x = \frac{\log \frac{3}{7}}{\log \frac{25}{3}}$

DISEQ. $3 \cdot 3^x = 3^{x+1}$

$3 \cdot 2^x > 4 \cdot 3^x$

$3 \cdot 2^x > 4 \cdot 3 \cdot 3^x$

$\frac{2^x}{2^2} > 3^x$; $2^{x-2} > 3^x$; $\ln 2^{x-2} > \ln 3^x$ (perché $x>1$)

$(x-2)\ln 2 > x \ln 3$

$x \ln 2 - 2\ln 2 > x \ln 3$

$x(\ln 2 - \ln 3) > 2\ln 2$ ($\ln 2 - \ln 3 < 0$)

$x < \frac{2\ln 2}{\ln 2 - \ln 3}$

564 $4 \cdot 3^x + 3^{x+1} = 2$

$4 \cdot 3^x + 3^x \cdot 3 = 2$

$3^x (4+3) = 2$

$3^x \cdot 7 = 2$

$3^x = \frac{2}{7}$

$\log 3^x = \log \frac{2}{7}$

$x \log 3 = \log 2 - \log 7$

$x = \frac{\log 2 - \log 7}{\log 3}$

569 $\frac{1}{2^x - 1} + \frac{2^x}{4^x - 1} = \frac{3 \cdot 2^x - 1}{2^x + 1}$

$2^x = t \rightarrow \frac{1}{t-1} + \frac{t}{t^2-1} = \frac{3t-1}{t+1} \rightarrow \frac{(t+1)+t-(t-1)(3t-1)}{(t-1)(t+1)} = 0$

C.E: $t \neq \pm 1 \rightarrow 2^x \neq 1 \rightarrow x \neq 0$

$t+1+t-3t^2+3t+t-1=0$

$3t^2 - 6t = 0 \rightarrow 3t(t-2) = 0$

$t_{1/2} \begin{cases} 0 = 2^x \rightarrow \text{impossibile} \\ 2 = 2^x \Leftrightarrow \boxed{x = 1} \end{cases}$

79 **LEGGI IL GRAFICO** Considera la funzione

$f(x) = a \log_2 |x + b| + c$

rappresentata nel grafico a fianco.

a. Determina i valori dei parametri a, b e c sulla base dei dati deducibili dal grafico.

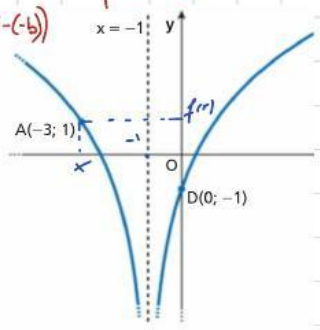
b. Risolvi la disequazione $f(x) \leq 3$.

c. Dimostra che $f(x)$ è invertibile nell'intervallo $]-1; +\infty[$ e scrivi l'espressione della funzione inversa.

[a) $a = 2, b = 1, c = -1$; b) $-5 \leq x < -1 \vee -1 < x \leq 3$;
c) $f^{-1}(x) = 2^{\frac{x+1}{2}} - 1$]

$\uparrow e \quad \leftarrow -b$

$y - e = f(x - (-b))$



- Asintoto $x = -1 \qquad (x+b) \neq 0 \rightarrow x \neq -b = -1 \Rightarrow b = 1$

$A \begin{cases} 1 = a \log_2 |-3+1| + c \\ -1 = a \log_2 |0+1| + c \end{cases} \Rightarrow \begin{cases} 1 = a \log_2 2 + c \\ -1 = a \log_2 1 + c \end{cases} \rightarrow \begin{cases} 1 = a + c \\ -1 = c \end{cases} \Rightarrow \begin{cases} a = 2 \\ c = -1 \end{cases}$

$f(x) = 2 \log_2 |x+1| - 1$

- $f(x) \leq 3 \Rightarrow 2 \log_2 |x+1| - 1 \leq 3$

$\log_2 |x+1| \leq 2$

$\log_2 |x+1| \leq \log_2 4$

$|x+1| \leq 4$

$x + 1 \leq 4 \qquad -x - 1 \leq 4$

$\begin{cases} x \leq 3 \\ x \geq -5 \end{cases}$



$-5 \leq x < -1 \vee -1 < x \leq 3$

$|x-a| \le b$ Intervallo centrato in a e raggio b

$\Downarrow$

$-b \le x-a \le b \Rightarrow a-b \le x \le a+b$

$|x+1| \le 4$





x

$f^{-1}(x) = 2\log_2(x+1) - 1$ (y)

$D: x > -1$

$x = 2\log_2(y+1) - 1$

$2\log_2(y+1) = \frac{x+1}{2}$

$2^{\frac{x+1}{2}} = y+1$

$y = 2^{\frac{x+1}{2}} - 1$

$f^{-1}(x) = 2^{\frac{x+1}{2}} - 1$

62 Considerate le funzioni $f(x) = |x-2|$ e $g(x) = \log_2 x - 2$,

a. esprimi $h = f \circ g$ e $t = g \circ f$;

b. rappresenta graficamente f, g, h e t;

c. risolvi le disequazioni: $h(x) > 1$, $t(x) > -2$.

[a) $h(x) = |\log_2 x - 4|$, $t(x) = \log_2|x-2| - 2$; c) $0 < x < 8 \vee x > 32$; $x < 1 \vee x > 3$]

$h = f \circ g$

$t = g \circ f$

$h(x) = f(g(x)) = |\log_2 x - 2 - 2| = |\log_2 x - 4|$

$t(x) = g(f(x)) = \log_2|x-2| - 2$

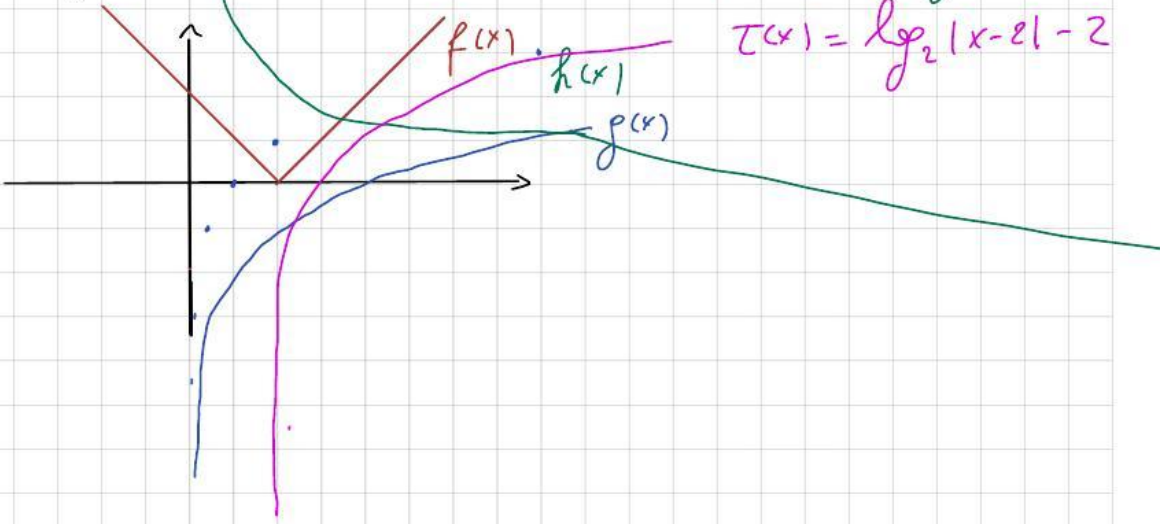
$f(x) = |x-2|$

$g(x) = \log_2 x - 2$

$h(x) = |\log_2 x - 4|$

$t(x) = \log_2|x-2| - 2$

x	$\log_2 x$
$\frac{1}{2}$	-1
1	0
2	1
8	3



$y = f(x)$

$y - b = f(x - a)$

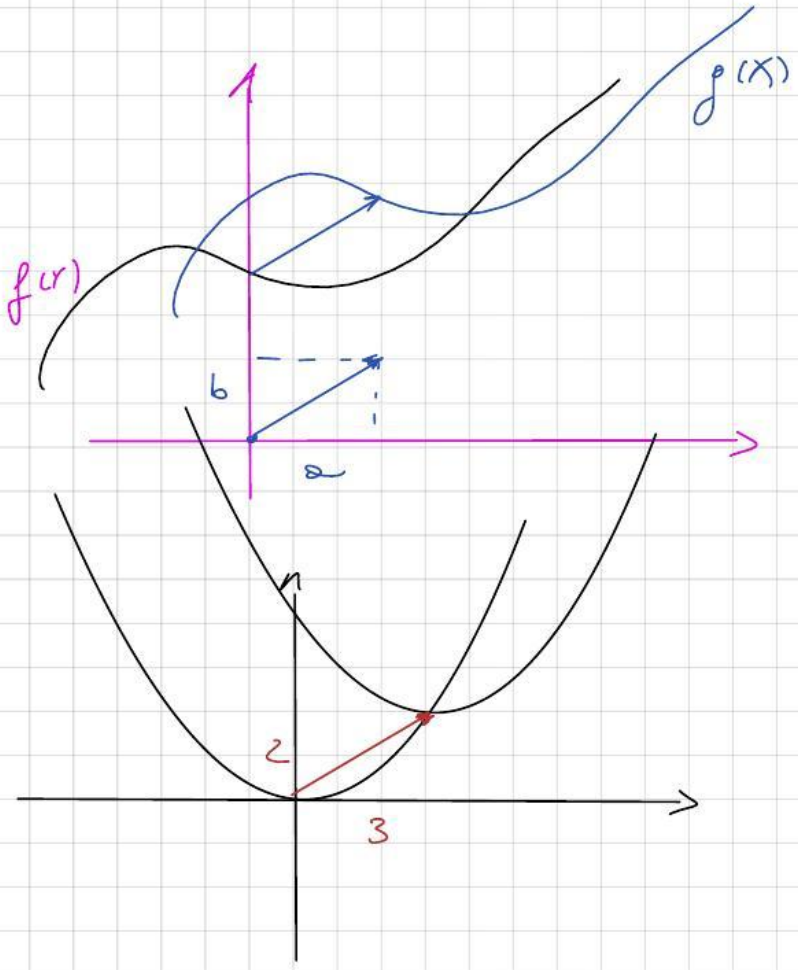
$y = f(x - a) + b$

$y = f(x - 2) + 3$

$y = x^2$

$y - 2 = (x - 3)^2$

$y = x^2 - 6x + 11$



80 Considera la funzione $f(x) = a\log_{\frac{1}{2}}(x+b)+c$.

a. Determina i valori di a, b e c tali che il grafico di $f(x)$ passi per il punto (2; 1), abbia la retta di equazione $x = 1$ come asintoto verticale e intersechi la retta di equazione $y = 2x - 3$ nel punto che ha in comune con la bisettrice del primo e terzo quadrante.

b. Rappresenta graficamente $f(x)$ applicando le trasformazioni geometriche e determina dominio, intersezioni con gli assi e segno della funzione trovata.

c. Risolvi l'equazione $f(x) = 5$.

d. Risolvi la disequazione $\dfrac{f(x)}{\sqrt{x-1}+x^2+4} < 0$.

[a) $a=-2, b=-1, c=1$; b) $D:]-1;+\infty[$; $\left(1+\frac{\sqrt{2}}{2};0\right)$; $f(x)$ positiva per $x > 1+\frac{\sqrt{2}}{2}$; c) $x = 5$; d) $\left]1;1+\frac{\sqrt{2}}{2}\right[$]

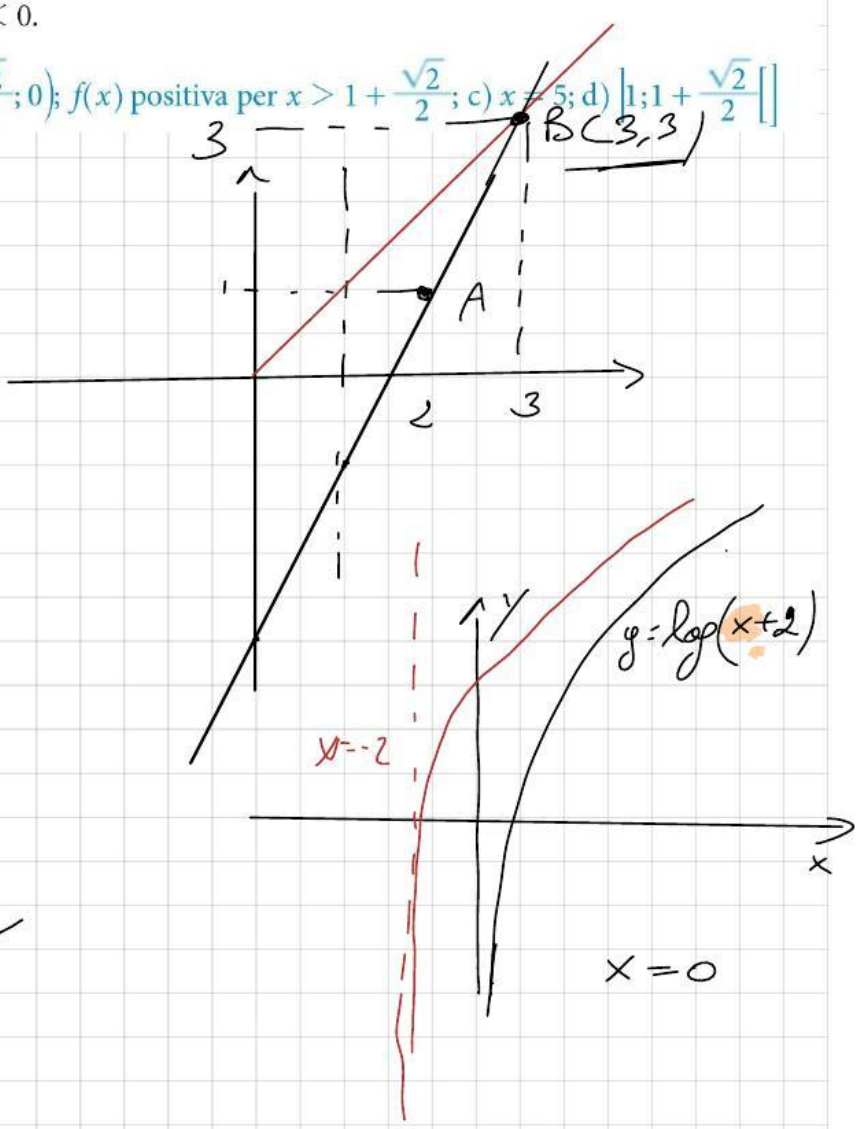
a) A(2,1)

$x = 1$ ASINTOTO

$y = 2x - 3$

$\begin{cases} y = x \\ y = 2x - 3 \end{cases}$ $\begin{cases} y = x \\ x = 2x - 3 \end{cases}$

$\begin{cases} y = x \\ x = 3 \end{cases}$ $\begin{cases} y = 3 \\ x = 3 \end{cases}$



$f(x) = a\log_{\frac{1}{2}}(x+b)+c$

$1+b=0 \rightarrow b=-1$

$\begin{cases} 1 = a\log_{\frac{1}{2}}(2-1)+c \\ 3 = a\log_{\frac{1}{2}}(3-1)+c \end{cases}$ $\begin{cases} 1 = a\cdot 0 + c \\ 3 = a(-1)+1 \end{cases}$ $\begin{cases} c = 1 \\ a = -2 \end{cases}$

$-\log_2(2)$

$\log^{[2]}(x) \neq \log x^{[2]}$

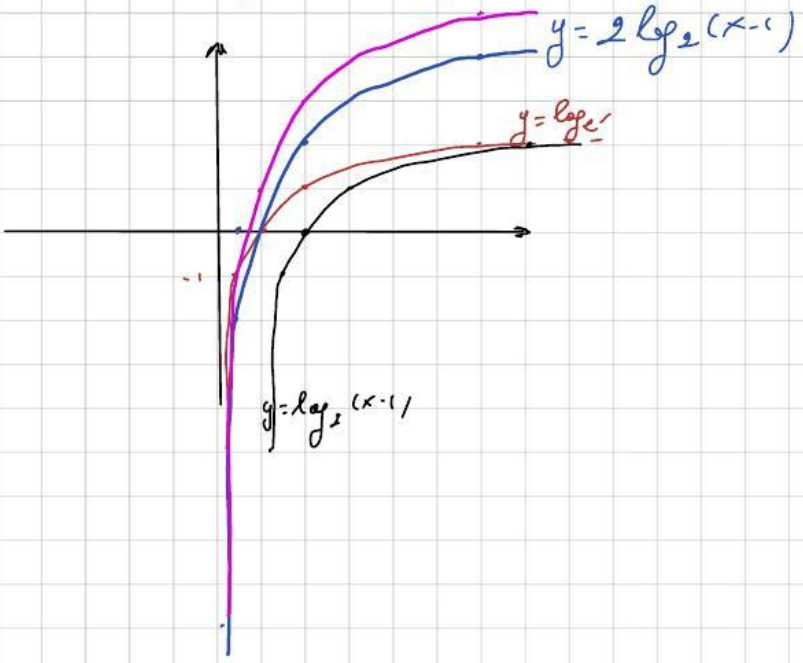
$$\log x \cdot \log x = (\log x)^2$$

$f(x) = -2\log_{\frac{1}{2}}(x-1)+1$

$f(x) = 2\log_2(x-1)+1$

$\log_{\frac{1}{a}} x = -\log_a x$

1) $y = \log_2 x$
2) $y = \log_2(x-1)$
3) $y = 2\log_2(x-1)$
4) $y = 2\log_2(x-1)+1$



DOMINIO:

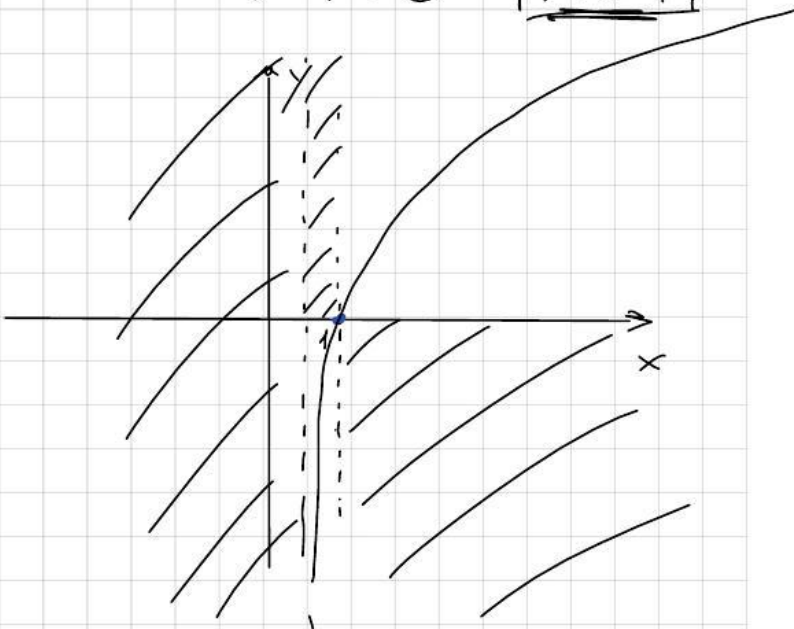
$f(x) = 2\log_2(x-1)+1$ $\quad x-1>0 \quad \boxed{x>1}$

$D: x>1$

$\mathcal{D}: (1, +\infty)$

$]1, +\infty[$

$D = \{x \in \mathbb{R} : x > 1\}$



Int.

Asse y $\begin{cases} x=0 \\ y = 2\log_2(x-1)+1 \end{cases}$

Asse x $\begin{cases} y=0 \\ y = 2\log_2(x-1)+1 \end{cases} \begin{cases} 0 = 2\log_2(x-1)+1 \end{cases}$

non ha p.d.i. con l'asse y.

$\log_2(x-1) = -\frac{1}{2}$ $\qquad 2^{-\frac{1}{2}} = (x-1) \Rightarrow x = \frac{\sqrt{2}}{2}+1 = \boxed{\frac{\sqrt{2}+2}{2}}$

segno: $y = 2\log_2(x-1)+1$

$2\log_2(x-1)+1 > 0$

$\log_2(x-1) > -\frac{1}{2}$; $\log_{>1}(x-1) > \log_{>1} 2^{-\frac{1}{2}}$; $x-1 > 2^{-\frac{1}{2}}$; $x > \frac{\sqrt{2}+2}{2}$

$f(x)$ è positive per $x > \frac{\sqrt{2}+2}{2}$

$f(x)$ è negative per $1 < x < \frac{\sqrt{2}+2}{2}$

21 $2\log_4^2|x+1| + \log_4|x^2-1| + \log_{\frac{1}{4}}|x-1| - 1 = 0$

C.E.

$(x+1) \neq 0$

$(x^2-1) \neq 0 \rightarrow (x+1)(x-1) \neq 0 \quad x \neq \pm 1$

$(x-1) \neq 0$

$2\log_4^2(x+1) + \log_4(|x+1|\cdot|x-1|) - \log_4|x-1| - 1 = 0$

$2\log_4^2|x+1| + \log_4|x+1| + \log_4|x-1| - \log_4|x-1| - 1 = 0$

$2t^2 + t - 1 = 0$

$t_{1/2} = \frac{-1 \pm \sqrt{1+8}}{4} = \frac{-1 \pm 3}{4}$ $\begin{cases} -1 = \log_4|x+1| \\ \frac{1}{2} = \log_4|x+1| \end{cases}$

$4^{-1} = |x+1| \rightarrow |x+1| = \frac{1}{4}$ $\begin{cases} x+1 = \frac{1}{4} \rightarrow x = -\frac{3}{4} & \text{se } x > -1 \\ -x-1 = \frac{1}{4} \rightarrow x = -\frac{5}{4} & \text{se } x < -1 \end{cases}$

$4^{1/2} = |x+1| \rightarrow |x+1| = 2$ $\begin{cases} x+1 = 2 \rightarrow x = 1 & x > -1 \\ -x-1 = 2 \rightarrow x = -3 & x < -1 \end{cases}$ non accettabile

-MAT: Stifani, Ruffilli, Innamorato

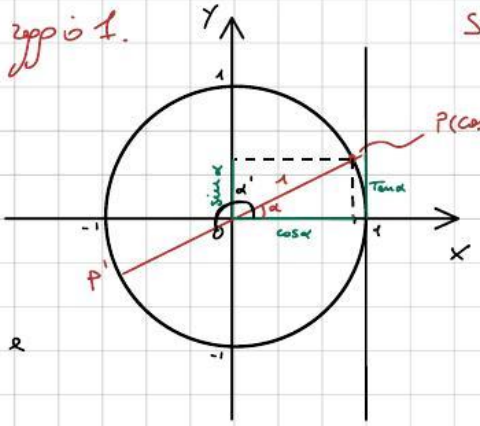
-FIS: Chiecchi Scalini

LE FUNZIONI GONIOMETRICHE

Relaz. goniom. fond.

$\sin^2\alpha + \cos^2\alpha = 1$

Circonferenza goniometrica: circ. di raggio 1.



	sin	cos
I	+	+
II	+	−
III	−	−
IV	−	+

Chiamiamo:

- Seno di α la proiezione di P sull'asse y
- Coseno di α la proiezione di P sull'asse x
- Tangente di α il rapporto tra seno e coseno

Il sin e il cos sono funzioni limitate:

$-1 \le \sin\alpha \le 1 \; ; \; -1 \le \cos\alpha \le 1$

$\tan\alpha = \frac{\sin\alpha}{\cos\alpha}$. Dim: applichiamo la proporzione ai Triangoli simili:

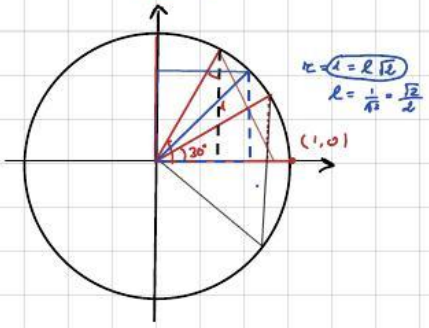
$\tan\alpha : 1 = \sin\alpha : \cos\alpha$

$\tan\alpha = \frac{\sin\alpha}{\cos\alpha}$

$\alpha' = \alpha + 180°$

$\alpha' - \alpha = 180°$

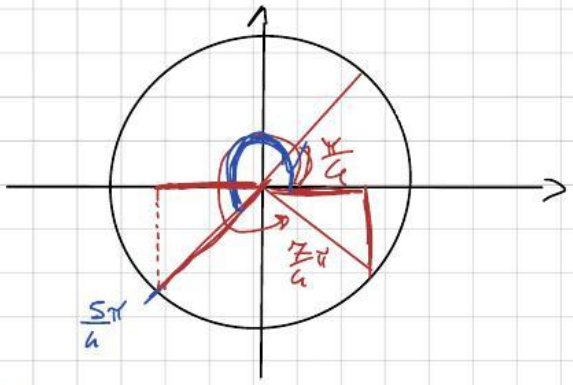
$\tan\alpha = \tan\alpha'$



$\alpha°$	α^c	$\sin\alpha$	$\cos\alpha$	$\tan\alpha$
$0°$	0	0	1	$\frac{0}{1} = 0$
$30°$	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
$45°$	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$60°$	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$90°$	$\frac{\pi}{2}$	1	0	∄

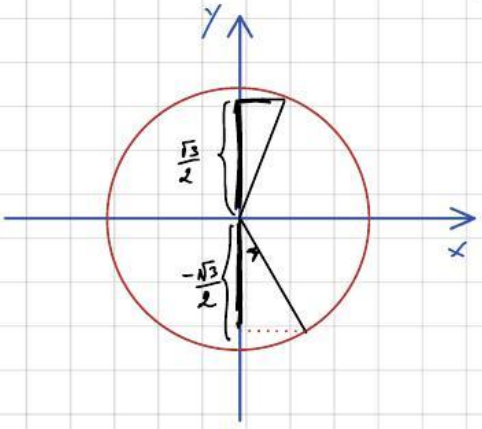
$\cos\left(\frac{7}{4}\pi\right)$

$\alpha = \frac{7}{4}\pi = 7 \cdot \frac{\pi}{4}$



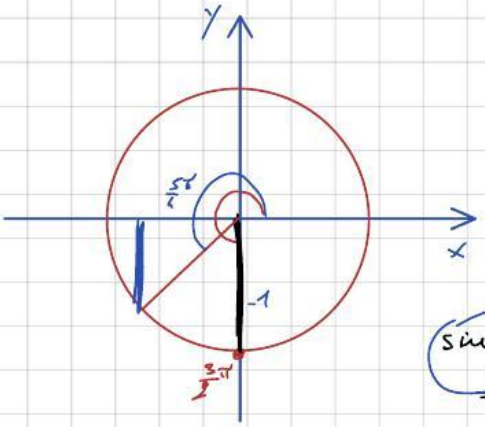
$\sin 300° = \sin(-60°)$

$\sin(5 \cdot 60°) \neq 5 \sin 60°$



$\frac{\sqrt{2}}{2} = 0{,}7071\ldots$

$\frac{\sqrt{3}}{2} = 0{,}866\ldots$



$\sin\frac{3}{2}\pi < \sin\frac{5}{4}\pi$

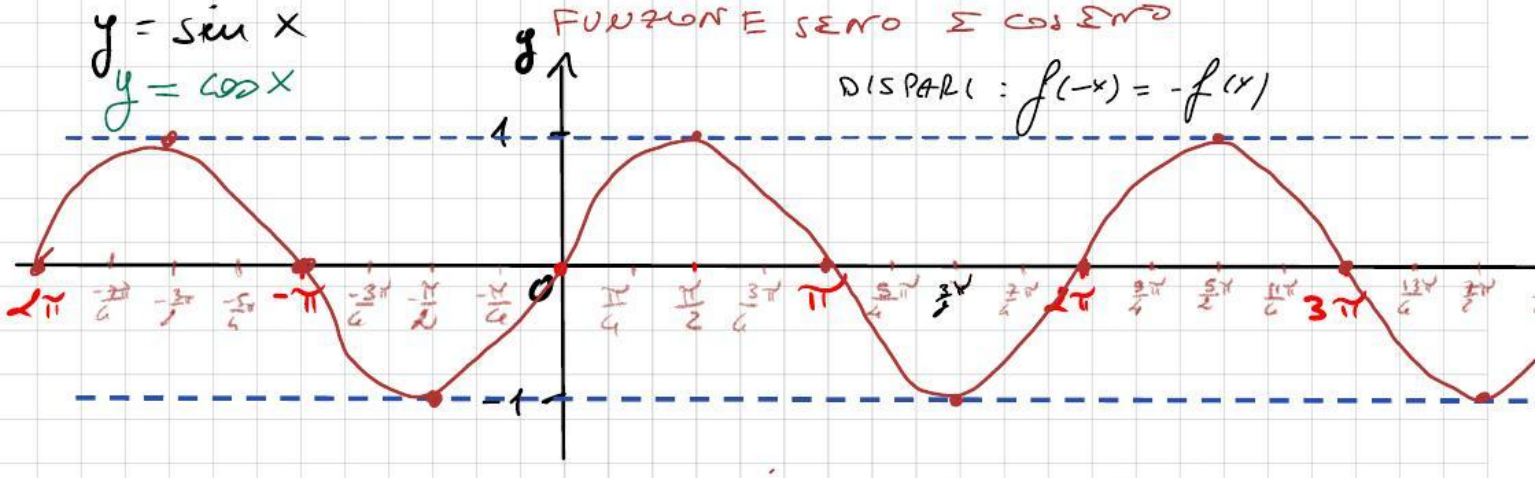
$-1 < -\frac{\sqrt{2}}{2}$

$y = \sin x$

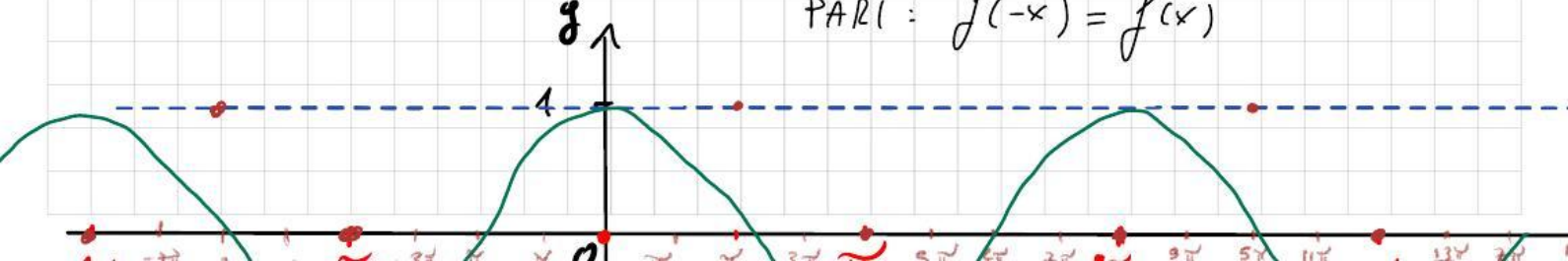
$y = \cos x$

FUNZIONE SENO E COSENO

DISPARI: $f(-x) = -f(x)$



PARI: $f(-x) = f(x)$

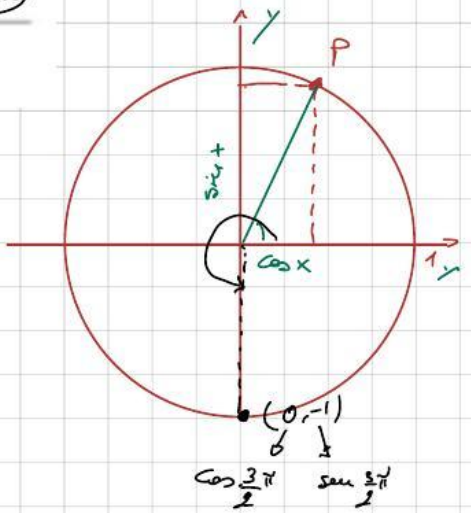


138 $\dfrac{(-\sin 5\pi + \cos\pi)\sin\frac{11}{2}\pi + 2\sin\frac{3}{2}\pi\cdot\cos 2\pi}{3\left(\cos\frac{5}{2}\pi + 2\sin\frac{9}{2}\pi\right)}$

$\sin 5\pi = 0$, $\cos\pi = -1$, $\cos 2\pi = 1$, $\cos\frac{5}{2}\pi = \cos\frac{\pi}{2}$

$$\frac{(0-1)\cdot(-1) + 2\cdot(-1)\cdot 1}{3(0+2\cdot 1)} = \frac{1-2}{6} = -\frac{1}{6}$$

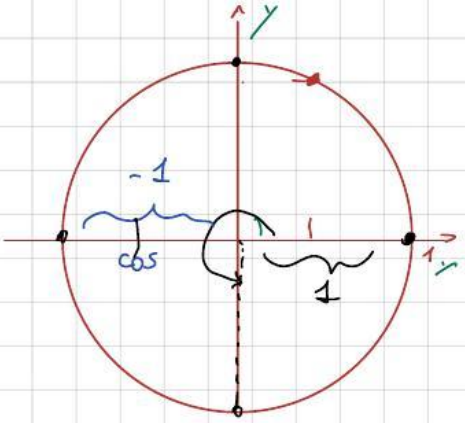
$$\frac{3}{2}\pi = \frac{8+1}{2}\pi = 4\pi + \frac{\pi}{2}$$



137 $\left(a\cos 2\pi + b\sin\frac{7}{2}\pi\right)^2 - \left[a\sin\left(-\frac{3}{2}\pi\right) + b\cos(-5\pi)\right]^2$

$(a\cdot 1 + b(-1))^2 - [a\cdot 1 + b(-1)]^2$

$(a-b)^2 - (a-b)^2 = 0$



$y = \sin x$ é ímpar

$y = \cos x$ é par

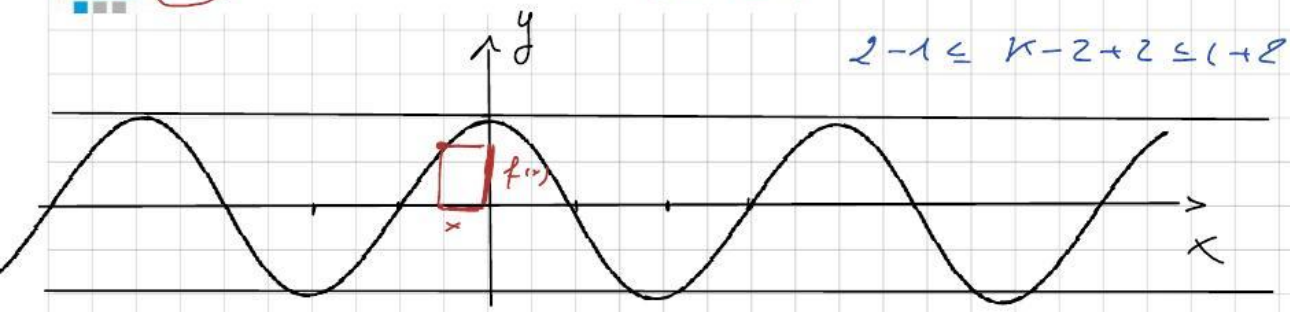
$\sin\left(-\frac{\pi}{2}\right) = -\sin\left(\frac{\pi}{2}\right)$

$\cos\left(-\frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2}\right)$

116 $\cos x = k - 2$ $\quad [1 \le k \le 3]$

$-1 \le k-2 \le 1$

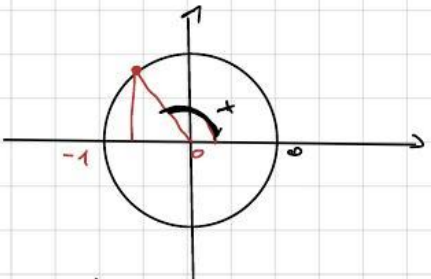
$2-1 \le k-2+2 \le 1+2$



$1 \le k \le 3$

121 $(2a-3)\cos x = -a+4$, con $x \in$ secondo quadrante.

$\cos x = \dfrac{-a+4}{2a-3}$ $\qquad 2a-3 \neq 0 \to a \neq \dfrac{3}{2}$



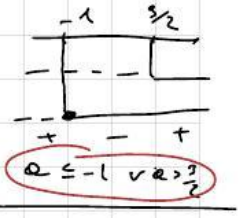
$-1 \le \dfrac{-a+4}{2a-3} \le 0$

$\begin{cases} \dfrac{+a-4}{2a-3} \le +1 \\ \dfrac{-a+4}{2a-3} \le 0 \end{cases} \to \begin{cases} \dfrac{a-4-2a+3}{2a-3} \le 0 \\ \dfrac{-a+4}{2a-3} \le 0 \end{cases} \to \begin{cases} \dfrac{-a-1}{2a-3} \le 0 \\ \dfrac{-a+4}{2a-3} \le 0 \end{cases}$

$\begin{cases} \dfrac{a+1}{2a-3} \ge 0 \\ \dfrac{-a+4}{2a-3} \le 0 \end{cases}$

N: $a+1 \ge 0 \to a \ge -1$

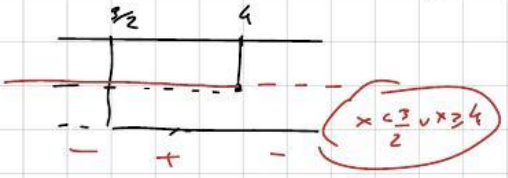
D: $2a-3 > 0 \to a > \dfrac{3}{2}$



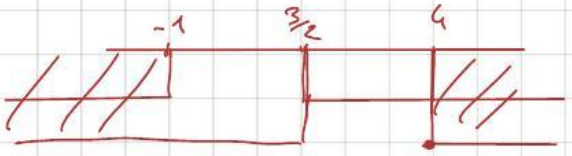
$a \le -1 \vee a > \dfrac{3}{2}$

N: $-a+4 \ge 0 \to a \le 4$

D: $a > \dfrac{3}{2}$



$x < \dfrac{3}{2} \vee x \ge 4$

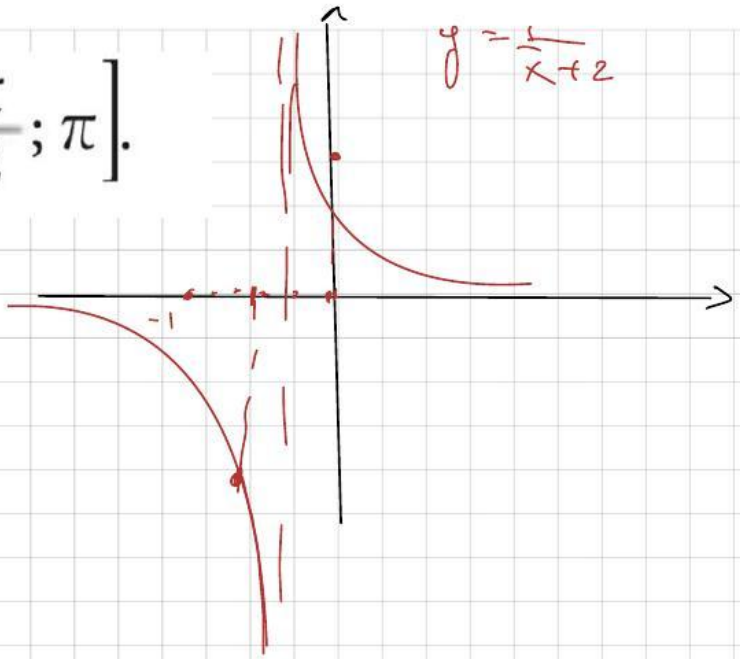


$a \le -1 \quad \vee \quad a \ge 4$

$$y = \frac{1}{2+\cos x}, \quad \left[\frac{\pi}{2}; \pi\right].$$

$[-1, 0]$

$y = \frac{1}{2+x}$



$y = \frac{1}{2-1} = 1$

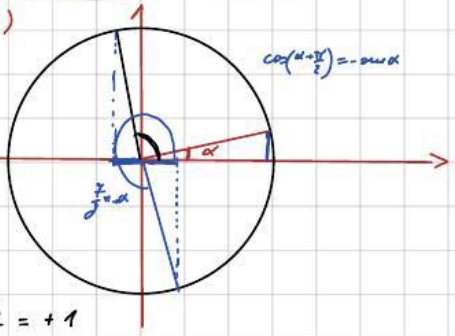
$y = \frac{1}{2+0} = \frac{1}{2}$

139 $\dfrac{a\sin\alpha + b\cos 2\alpha}{\sin(-4\alpha) - a\cos\left(\alpha+\frac{\pi}{2}\right) - b\cos\left(\frac{7}{2}\pi+\alpha\right)}$, con $\alpha = \frac{\pi}{2}$.

$\neq (-)$

$\cos\left(\alpha+\frac{\pi}{2}\right) = -\sin\alpha$

$\frac{a\sin\alpha + b\cos 2\alpha}{\sin(-4\alpha) - (a-b)\cos\left(\alpha+\frac{\pi}{2}\right)}$ ($-\sin\alpha$)



$\frac{a\sin\alpha + b\cos 2\alpha}{\sin(-4\alpha) + (a-b)\sin\alpha}$

$\frac{a\sin\left(\frac{\pi}{2}\right) + b\cos\pi}{-\sin(2\pi) + (a-b)\sin\frac{\pi}{2}} = \frac{a-b}{-b+a} = +1$

($\sin(2\pi) = 0$; $\sin\frac{\pi}{2} = 1$)

COTANGENTE

$\cot\alpha = \frac{\cos\alpha}{\sin\alpha} = \frac{1}{\tan\alpha}$

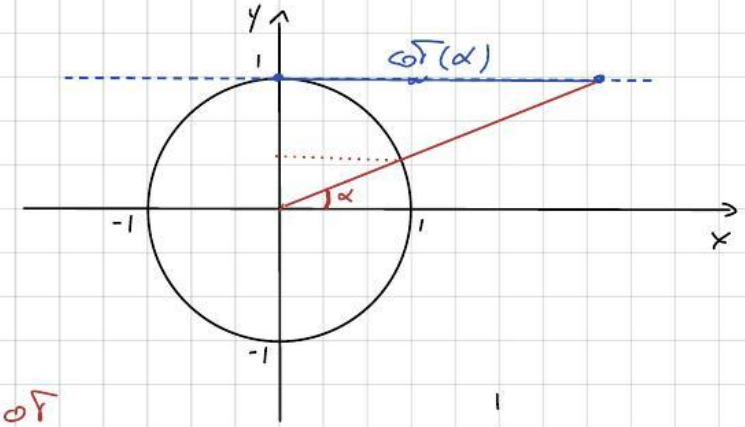
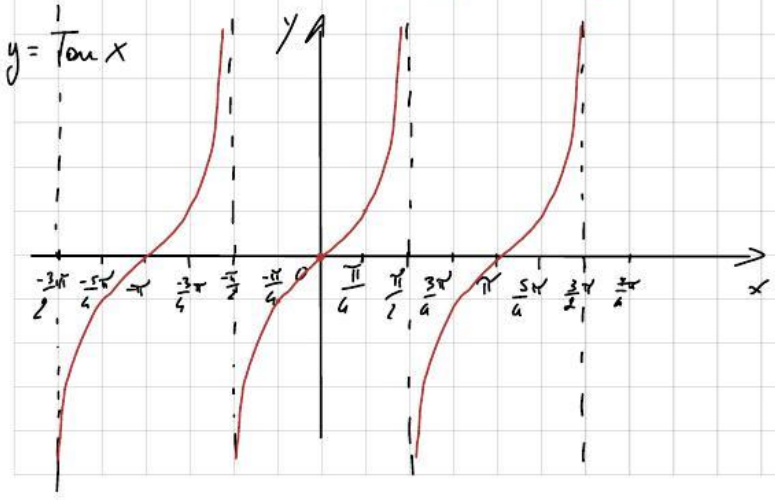
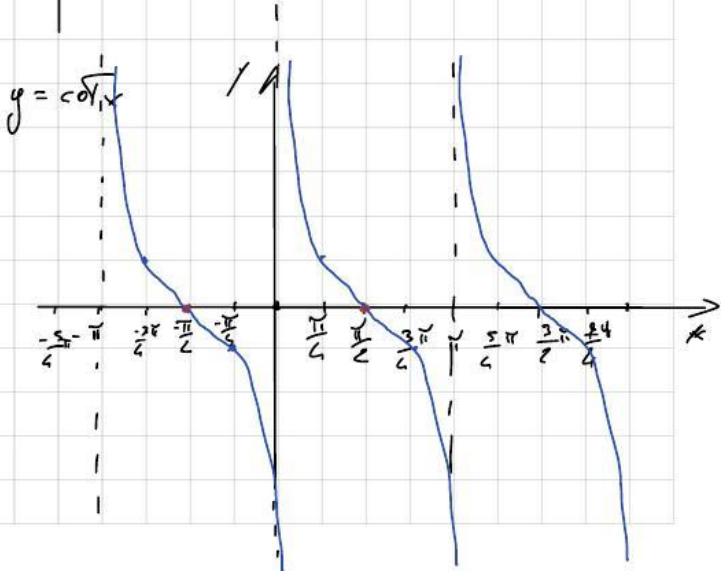


GRAFICO DELLA TAN e COT





PERIODICITÀ

$y = \sin x$ $\quad T = 2\pi$

$y = \cos x$ $\quad T = 2\pi$

$y = \tan x$ $\quad T = \pi$

$y = \cot x$ $\quad T = \pi$

$y = \sin 2x$ $\quad T = \pi$

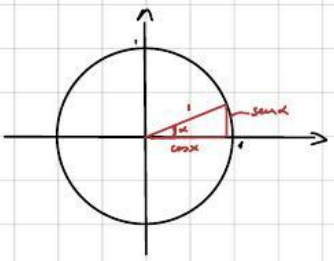
$y = \sin kx$

$y = \cos kx$

$T = \frac{2\pi}{k}$

n° 155

$\sin\alpha = \frac{4}{25}$ $\quad 0 < \alpha < \frac{\pi}{2}$ $\quad \cos\alpha = ?$



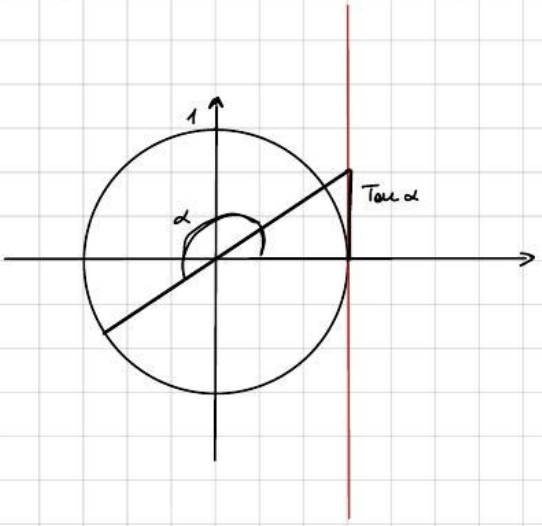
$\sin^2\alpha + \cos^2\alpha = 1$

$\cos^2\alpha = 1 - \sin^2\alpha = 1 - \left(\frac{4}{25}\right)^2 = 1 - \frac{16}{125} = \frac{625-16}{625} = \frac{576}{625} = \frac{24}{25}$

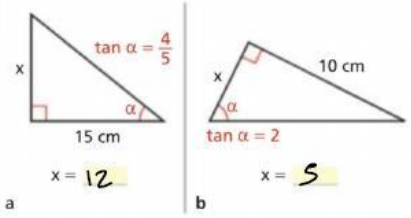
$\cos\alpha = + \frac{24}{25}$

• n° 181

$\tan\alpha = \frac{2}{3}$ $\quad$ asintoti



188

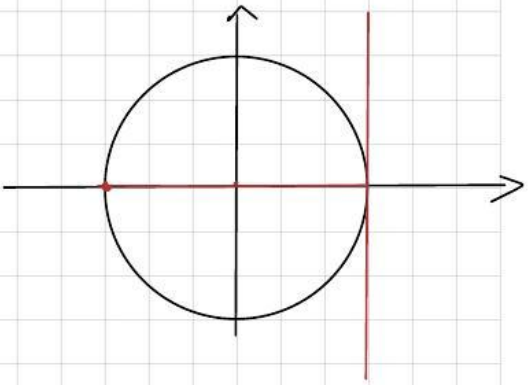


$\tan\alpha = \frac{\text{cateto opposto}}{\text{cateto adiacente}} = \boxed{\frac{x}{15} = \frac{4}{5}}$ $\quad x = \frac{15}{5} \cdot 4 = 12$

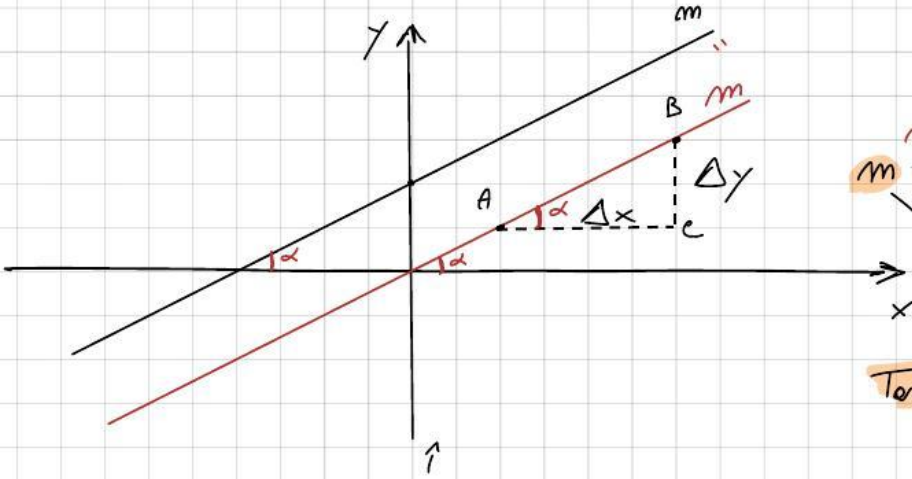
201 $\left(3\underbrace{\cos\pi}_{-1} + 2\underbrace{\sin 2\pi}_{0} + \underbrace{\tan 3\pi}_{0}\right)\left(8\underbrace{\sin\frac{\pi}{2}}_{1} - 2\underbrace{\cos 2\pi}_{1} + \frac{1}{2}\underbrace{\tan 8\pi}_{0}\right)$

$(3\cdot(-1) + 0 + 0)\,(8\cdot 1 - 2\cdot 1 + 0)$

$-3 \cdot 6 = -18$



SIGNIFICATO GEOMETRICO DEL COEFF. ANGOLARE



$y = mx + q$

$m = \frac{\Delta y}{\Delta x} = \frac{y_B - y_A}{x_B - x_A}$

$\tan\alpha = \frac{\overline{BC}}{AC} = \frac{\Delta y}{\Delta x}$

$\tan\alpha = \frac{1}{2} \rightarrow \alpha = \arctan\frac{1}{2} = 26{,}5°$

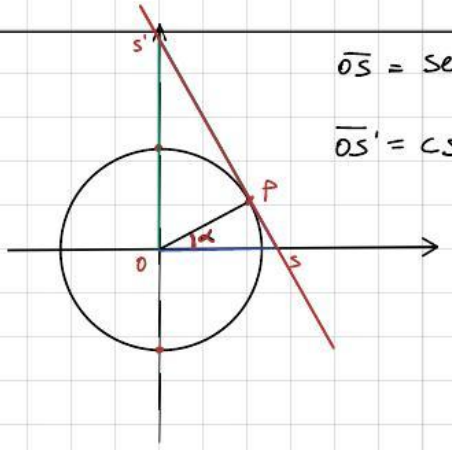
SECANTE

$$\sec\alpha = \frac{1}{\cos\alpha}$$

con $\alpha \neq \frac{\pi}{2} + k\pi$ con $k \in \mathbb{Z}$

$$\csc\alpha = \frac{1}{\sin\alpha}$$

con $\alpha \neq k\pi$ con $k \in \mathbb{Z}$



$\overline{OS} = \sec\alpha$

$\overline{OS'} = \csc\alpha$

Troviamo gli zeri di $\cos\alpha$.

$\alpha_1 = \frac{\pi}{2}$; $\alpha_2 = \frac{3}{2}\pi$; $\alpha_3 = \frac{5}{2}\pi$; $\alpha_4 = \frac{7}{2}\pi$ ....

$\alpha = \frac{\pi}{2} + k\pi$

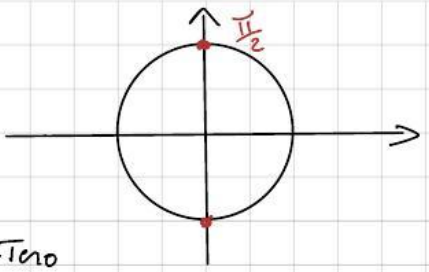
$k = 0 \rightarrow \frac{\pi}{2} = \alpha_1$

$k = 1 \rightarrow \frac{\pi}{2} + \pi = \frac{3}{2}\pi$

$k = 2 \rightarrow \frac{\pi}{2} + 2\pi = \frac{5}{2}\pi$

DOMINIO:

146 $y = \dfrac{\sqrt{1+\sin x}}{\cos x}$ $\left[x \neq \dfrac{\pi}{2} + k\pi\right]$



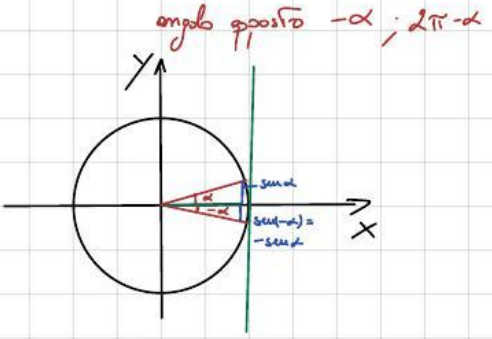
C.E.: $\begin{cases} \cos x \neq 0 \\ 1 + \sin x \geq 0 \end{cases}$ $\begin{cases} x \neq \frac{\pi}{2} + k\pi \\ \forall x \in \mathbb{R} \end{cases}$ k intero

$\sin x \geq -1$

ANGOLI ASSOCIATI

360 $-\sin\alpha \sin(-\alpha) + \cos\alpha \cos(-\alpha)$

361 $\tan(-\alpha) - 2\cos(-\alpha) + 2\sin(-\alpha)\cot(-\alpha)$



360)

- $-\sin\alpha(-\sin\alpha) + \cos\alpha \cdot \cos\alpha =$

$\sin^2\alpha + \cos^2\alpha = 1$

- 361) $-\tan\alpha - 2\cos\alpha + 2\sin\alpha\cot\alpha$

$-\tan\alpha - 2\cos\alpha + 2\sin\alpha\dfrac{\cos\alpha}{\sin\alpha} = -\tan\alpha$

$\tan\alpha = \dfrac{\sin\alpha}{\cos\alpha}$ $\dfrac{D}{P} = D$

$\tan(-\alpha) = -\tan\alpha$

SPECCHIETTO SUCCA PARITÀ'

– Operazioni con funzioni simmetriche

SOMMA $h(x) = f(x) \pm g(x)$ $\quad D \;\dot{=}\; D \;\dot{=}\; D$ $\quad Es: y = x^3 + \sin x$

$h(-x) = (-x)^3 + \sin(-x) = -x^3 - \sin x = -(x^3 + \sin x) = -h(x)$

$D \pm D = D$

SOMMA $h(x) = f(x) \pm g(x)$ $\quad N.P.$ $\quad D \;\dot{=}\; P$ $\quad Es: y = \sin x + x^2$

$h(-x) = \sin(-x) + (-x)^2 = -\sin x + x^2 \neq -h(x)$

SOMMA $h(x) = f(x) \pm g(x) \Rightarrow x^2 + \cos x$ $\quad P \;\dot{=}\; P \;\dot{=}\; P$

$h(-x) = (-x)^2 + \cos(-x) \Rightarrow x^2 + \cos x = h(x)$

PROD. $h(x) = f(x) \cdot g(x) \Rightarrow y = x^3 \cdot \sin x$ $\quad P \;\dot{=}\; D \;\dot{=}\; D$

$h(-x) = (-x)^3 \cdot \sin(-x) = +x^3 \cdot \sin x = h(x)$

PROD $h(x) = f(x) \cdot g(x) \Rightarrow y = x^2 \cdot \sin x$ $\quad D \;\dot{=}\; P \;\dot{=}\; D$

$h(-x) = (-x)^2 \cdot \sin(-x) = -x^2 \cdot \sin x = -h(x)$

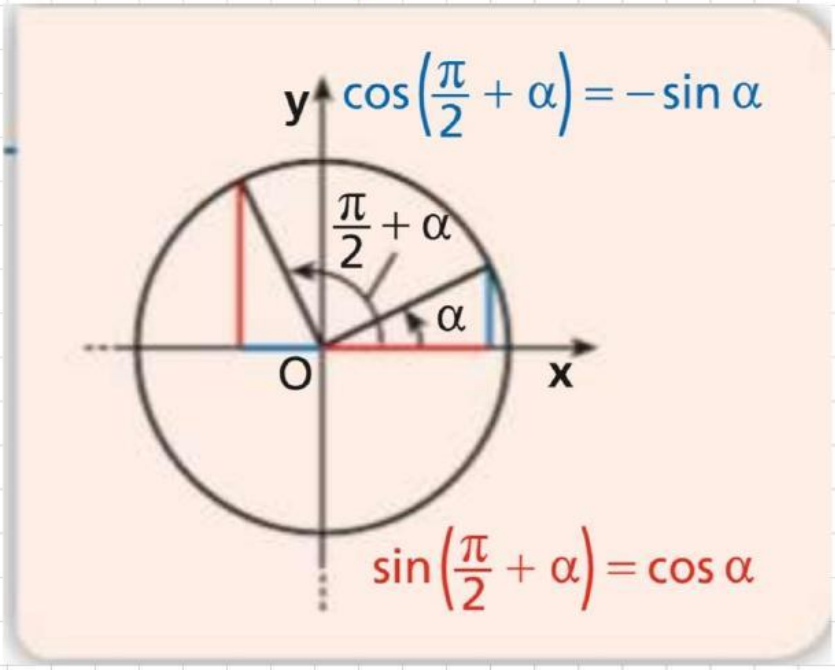
P P P

REACT $h(x) = f(x) \cdot f(x) \Rightarrow y = \cos x \cdot x^2$

$h(-x) = \cos(-x) \cdot (-x)^2 = \cos x \cdot x^2 = h(x)$

P + P = P
P + D = N.T
D + D = D
P · P = P
P · D = D
D · D = P

ANGOLI CHE DIFF. DI $\frac{\pi}{2}$



$\sin\left(\frac{\pi}{2}+\alpha\right) = \cos\alpha$

$\cos\left(\frac{\pi}{2}+\alpha\right) = -\sin\alpha$

383 $\sec\left(\frac{\pi}{2}+\alpha\right)\tan\alpha + \sin\left(\frac{\pi}{2}+\alpha\right) + \csc\left(\frac{\pi}{2}+\alpha\right)$

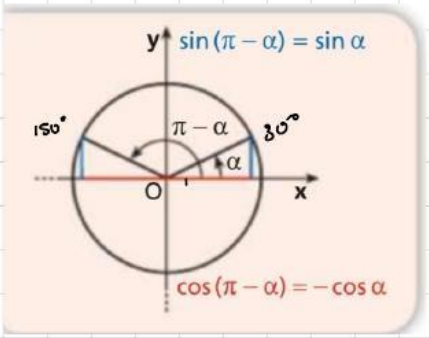
$\sec\alpha = \frac{1}{\cos\alpha}$ $\qquad$ $\csc\alpha = \frac{1}{\sin\alpha}$

$$\frac{1}{\cos\left(\frac{\pi}{2}+\alpha\right)} \cdot \frac{\sin\alpha}{\cos\alpha} + \sin\left(\frac{\pi}{2}+\alpha\right) + \frac{1}{\sin\left(\frac{\pi}{2}+\alpha\right)}$$

$$\frac{1}{-\sin\alpha} \cdot \frac{\sin\alpha}{\cos\alpha} + \cos\alpha + \frac{1}{\cos\alpha}$$

$$-\frac{1}{\cos\alpha} + \cos\alpha + \frac{1}{\cos\alpha} = \cos\alpha.$$

ANGOLI SUPPLEM.



$\sin(\pi-\alpha) = \sin\alpha$

$\cos(\pi-\alpha) = -\cos\alpha$

388 $2[\sin\alpha \sin(180°-\alpha) - \cos\alpha \cos(180°-\alpha)] - 5\cos 180°$

($\pi - \alpha$; -1)

$2[\sin\alpha \cdot \sin\alpha + \cos\alpha \cdot \cos\alpha] - 5 \cdot (-1)$

$\sin^2\alpha + \cos^2\alpha = 1$

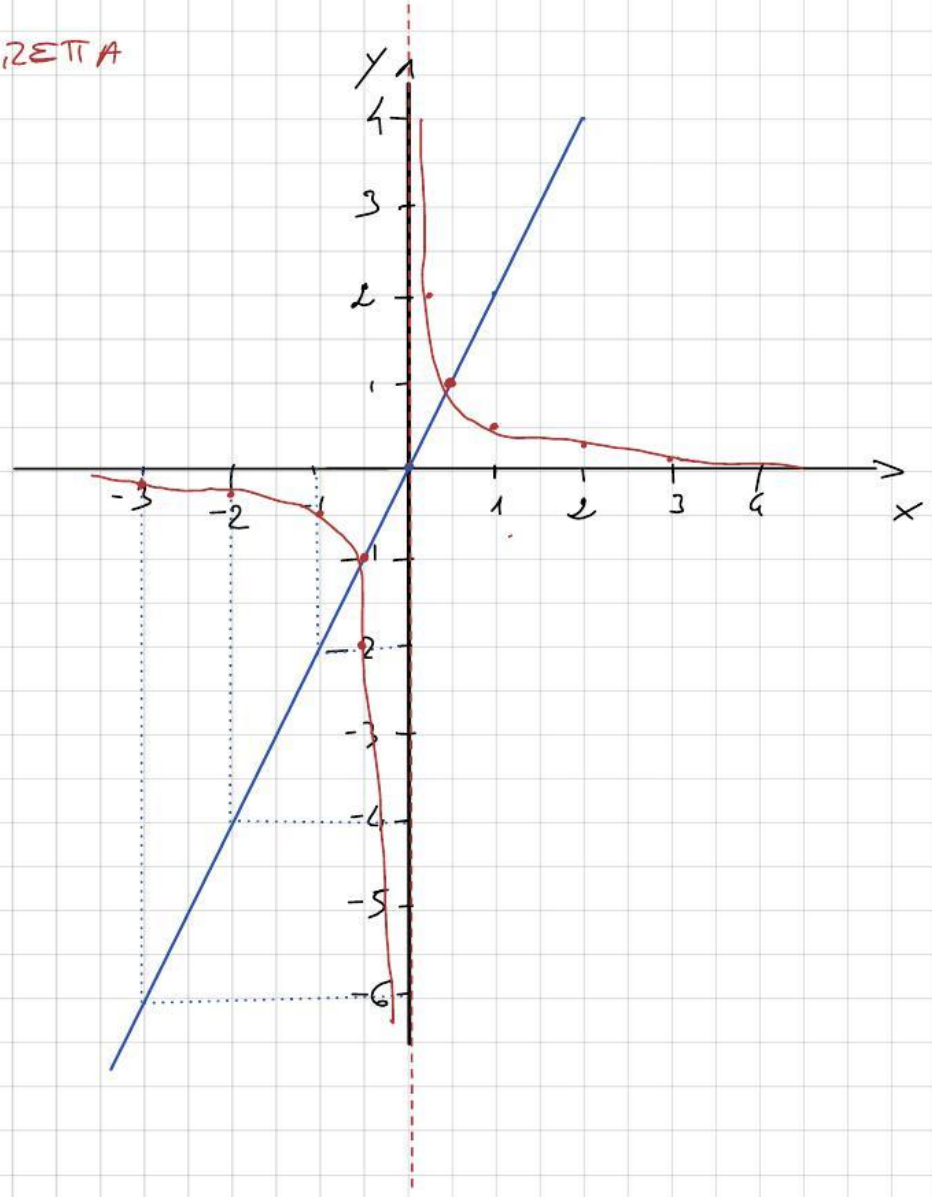
$2 + 5 = 7.$

IL RECIPROCO DELLA RETTA

$y = 2x$ (disegno)

$y' = \frac{1}{y} = \frac{1}{2x}$

x	y	y'
-3	-6	-1/6
-2	-4	-1/4
-1	-2	-1/2
-1/2	-1	-1
-1/4	-1/2	-2
0	0	∄
1/4	1/2	2
1/2	1	1
1	2	1/2
2	4	1/4
3	6	1/6



Il reciproco di una funzione

Il reciproco di una funzione $y = f(x)$ si ottiene calcolando l'inverso di ogni sua ordinata:

$$y' = \frac{1}{y} = \frac{1}{f(x)}$$

Per disegnare il grafico del reciproco partendo dalla funzione originale (nel tuo caso la retta blu $y = 2x$), basta seguire queste semplici regole:

- **Il segno non cambia:** Dove la funzione originale è positiva (sopra l'asse x), anche il suo reciproco sarà positivo. Dove è negativa (sotto l'asse x), il reciproco resterà negativo.
- **Comportamento opposto (crescente/decrescente):** Quando i valori di y aumentano, i valori del reciproco y' diminuiscono, e viceversa.
- **Punti fissi ($y = 1$ e $y = -1$):** I punti in cui la funzione vale 1 o -1 restano identici, perché $\frac{1}{1} = 1$ e $\frac{1}{-1} = -1$. Nel grafico, sono i punti in cui la curva rossa e la retta blu si incrociano.
- **Zeri e asintoti verticali ($y = 0$):** Nei punti in cui la funzione originale tocca l'asse x ($y = 0$), il reciproco **non esiste** perché non si può dividere per zero ($\frac{1}{0}$ è impossibile). In corrispondenza di questi punti si crea un **asintoto verticale** (la linea tratteggiata rossa).
- **Valori molto piccoli e molto grandi:**
  - Se y si avvicina moltissimo a 0, il suo reciproco y' diventa grandissimo (tende a $+\infty$ o $-\infty$, avvicinandosi all'asintoto verticale).
  - Se y diventa grandissimo (tende a $+\infty$ o $-\infty$), il suo reciproco y' si avvicina sempre di più a 0 (schiacciandosi verso l'asse x, che diventa un **asintoto orizzontale**).